Universiteit van Amsterdam

Social Network Analysis

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20. April 2026

Structure of this Lecture

1. Questions you can answer with SNA
2. A brief history of social networks
3. Is there a theory of social networks?
4. Network basics: terms, foundations and examples
5. Visual Network Analysis
6. Community detection
7. Case studies: (a) Economic mobility (b) Zwarte Piet Debate on Twitter

Three types of data in the social sciences:

1. Attribute data:

E.g., survey, demographics

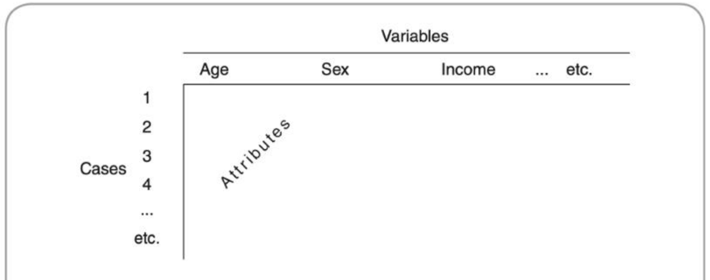
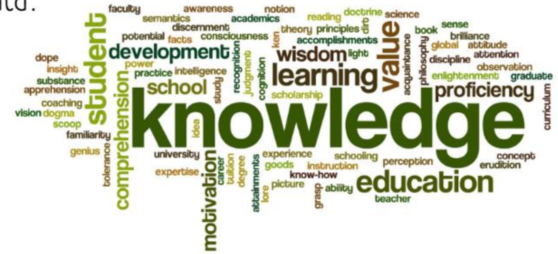


Figure 4.1 A data matrix for variable analysis

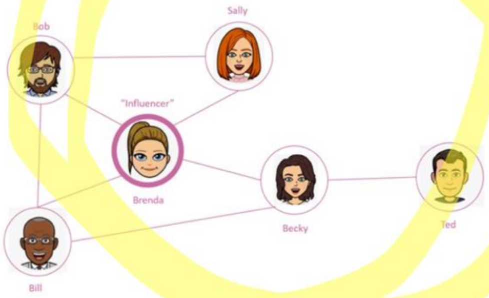
2. Ideational data:

E.g., text, speech

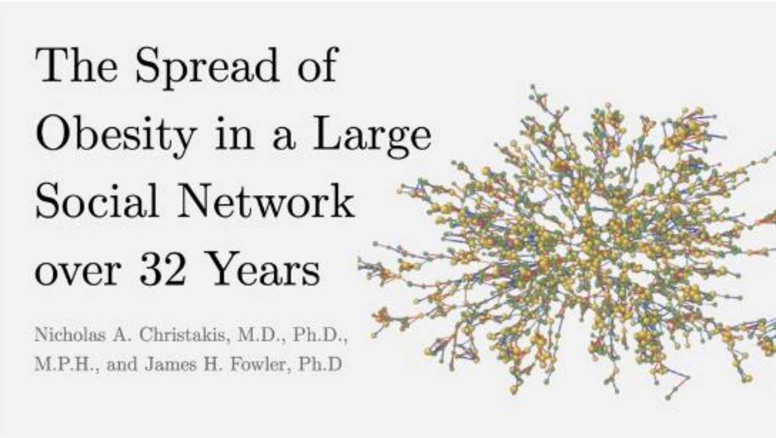


3. Relational data:

e.g. friendships, connections, etc.



Is Obesity Contagious? (Christakis & Fowler, 2007)



Data:
Framingham Heart Study, a long-term cardiovascular study initiated in 1948, which tracks the health and lifestyle of participants and their families over multiple generations. Longitudinal over 32 years.

Network:
Social networks based on familial, friendship, and spousal relationships within the Framingham cohort.

Finding:
If a person became obese, their friends, siblings, and even distant social contacts were more likely to become obese as well.

Obesity tends to "cluster" in social groups. If an individual's close friend became obese, the person was 57% more likely to become obese themselves. This likelihood increased if multiple people in the network were obese.
Obesity is not merely correlated with social relationships, but it spreads contagiously through social networks.

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7. 6 degrees of separation and the strength of weak ties
8. Case study: Zwarte Piet debate on Twitter

What questions can you answer with SNA?

How do friendship networks evolve over time in schools?

Who are the information brokers in the spread of innovation within a sector?

What are the patterns of professional networking that lead to successful job placement?

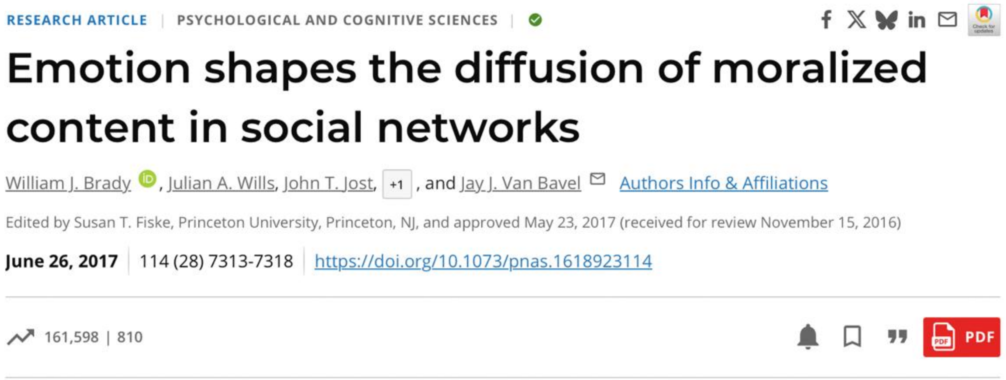
Which influencer should you buy if you want to maximize your social media viral ads in a particular social group?

Who is the central person within your friendship community?

What questions can you answer with SNA?

- **Centrality Analysis:** Identifies the most important or influential individuals within a network.
- **Community Detection:** Unearths the structure of the network by identifying clusters or communities within it.
- **Network Dynamics and Change Over Time:** Examines how social networks evolve, and who forms ties with whom.
- **Structural Holes and Network Brokers**: Analyzes the gaps in a social network (structural holes) and identifies brokers who bridge these gaps.
- **Network Cohesion and Subgroup Analysis:** Studies the strength of relationships within a network to identify tight-knit groups or subgroups. This can help in understanding the social cohesion or fault lines within a community.
- **Role Analysis:** Looks at the roles individuals play within a network based on their relationships and interactions. It can identify roles like leaders, gatekeepers, or peripheral members.
- **Diffusion of Information and Contagion:** Examines how information, behaviors, or innovations spread through a network. This is particularly relevant for understanding phenomena like viral marketing, spread of rumors, or adoption of new technologies.
- **Social Capital Analysis:** Explores the resources available to individuals through their social networks. It can shed light on how social ties contribute to personal or organizational success.

Example: Information Spread Online


RESEARCH ARTICLE | PSYCHOLOGICAL AND COGNITIVE SCIENCES |

Emotion shapes the diffusion of moralized content in social networks

William J. Brady, Julian A. Wills, John T. Jost, +1, and Jay J. Van Bavel Authors Info & Affiliations

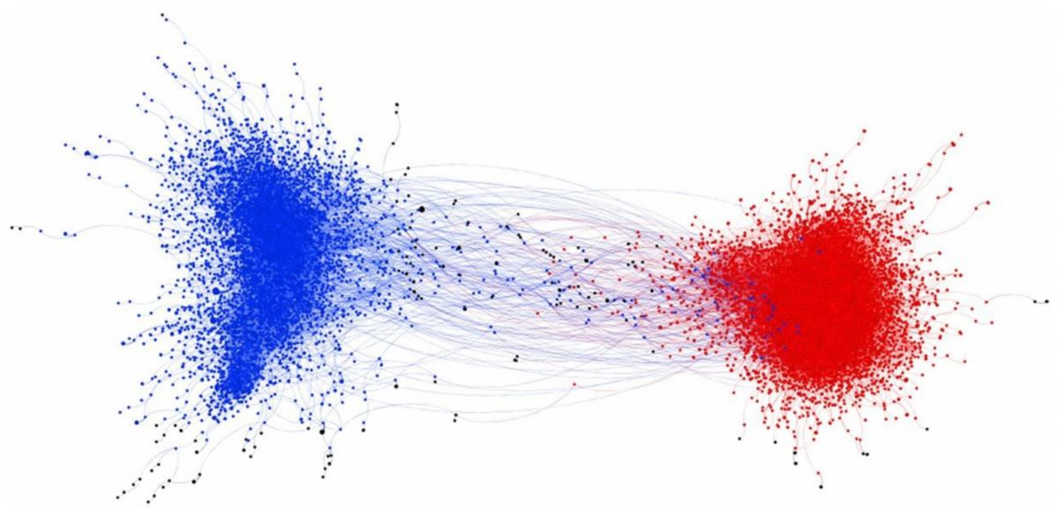
Edited by Susan T. Fiske, Princeton University, Princeton, NJ, and approved May 23, 2017 (received for review November 15, 2016)

June 26, 2017 | 114 (28) 7313-7318 | https://doi.org/10.1073/pnas.1618923114

161,598 | 810

PDF

- Combining NLP and Network analysis
- N=536,312 social media messages about three polarizing moral/political issues
- The presence of "*moral-emotional words*" in messages increased their diffusion by a factor of 20% for each additional word.
- Potential in-group advantage for *moral contagion:* wider spread of moral-emotional language within in-group networks than within out-group networks (right)



Network graph of moral contagion shaded by political ideology. The graph represents a depiction of messages containing moral and emotional language, and their retweet activity, across all political topics (gun control, same-sex marriage, climate change). Nodes represent a user who sent a message, and edges (lines) represent a user retweeting another user. The two large communities were shaded based on the mean ideology of each respective community (blue represents a liberal mean, red represents a conservative mean).

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A Brief History of Social Networks

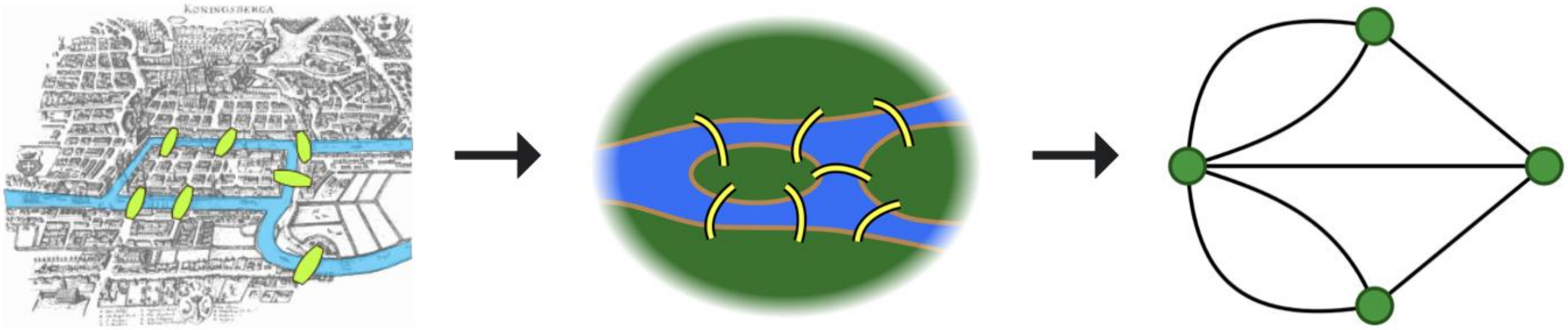
- Graph == network. They're just different words for the same thing, but “graph” is used in math, and “network” in more applied areas.
- Graph theory: **Leonhard Euler** in 1736.
- **The Königsberg Bridge problem**: Can you walk through the town in such a way that each bridge would be crossed exactly once?
- Euler:
  - Nope.
  - To solve it, he laid the foundations for a new mathematical structure: **the graph**

Graphs

Euler's approach to the Königsberg Bridge Problem:

Land mass = vertex/node

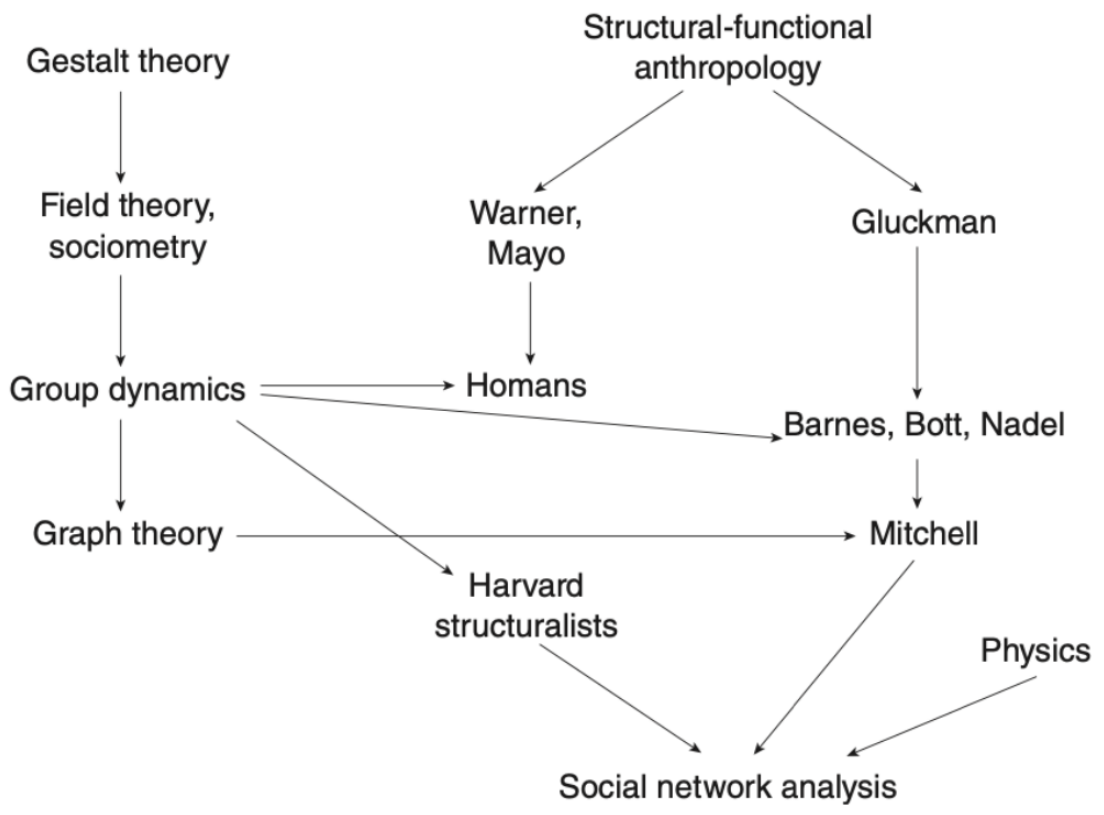
Bridges = edge (which pair of land masses is connected by a bridge?)



- Graphs are used in computer science, linguistics, physics, chemistry, biology
- *Social* networks = social science graphs

Origins & Developments of SNA in the Social Sciences

- **Early Developments:** Sociologists, anthropologists, psychologists explored social relationships (Early 20th Century). Georg Simmel, Émile Durkheim emphasized studying relationship patterns.
- **Formalization:** Introduction of sociograms by Jacob Moreno (1930s).
- **Growth Period:** Theoretical advancements by Harrison White and others (1940s-1970s).
- **Systematic Analysis Expansion:** Contributions by Ronald Burt, Barry Wellman, and others (1970s onwards).
- **Recent digital explosion:** Expansion due to computers and the internet; application in economics, epidemiology, political science. Advancements by Duncan J. Watts, Albert-Lászlo ́ Barabási



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Is There a Theory of Social Networks?

Several theories and movements align with SNA lens:

Theory of social capital: Robert Putnam 2000 “Bowling alone”. But reduces networks to individual value

Actor-Network Theory. Bruno Latour 2005. “actors” are not to be equated with human individuals or even groups but are to be seen as constituted by the relations that connect individuals to material objects

Manifesto for relational sociology. Emirbayer 1997: we should focus less on sociology as people with attributes, and more as a process of relationships.

Is There a Theory of Social Networks?Not really!

- Social networks cannot be reduced to a single theory
- They are a general set of techniques with many theoretical interpretations.
- A form of data or “a way of seeing”.
- Since they can be applied to very different things, we must always add an interpretation for what the measures and representation means!

Why SNA?

- Relationships and how we connect with one another matter!
- The way networks are patterned and structured also matters
- Surveys assume independence, but we are not independent; we are social beings.
- “The whole is more than the sum of its parts”. Nonlinearity. Complexity

"For the last thirty years, empirical social research has been dominated by the sample survey. But as usually practiced, ... the survey is a sociological meat grinder, tearing the individual from his social context and guaranteeing that nobody in the study interacts with anyone else in it."

- Allen Barton, 1968

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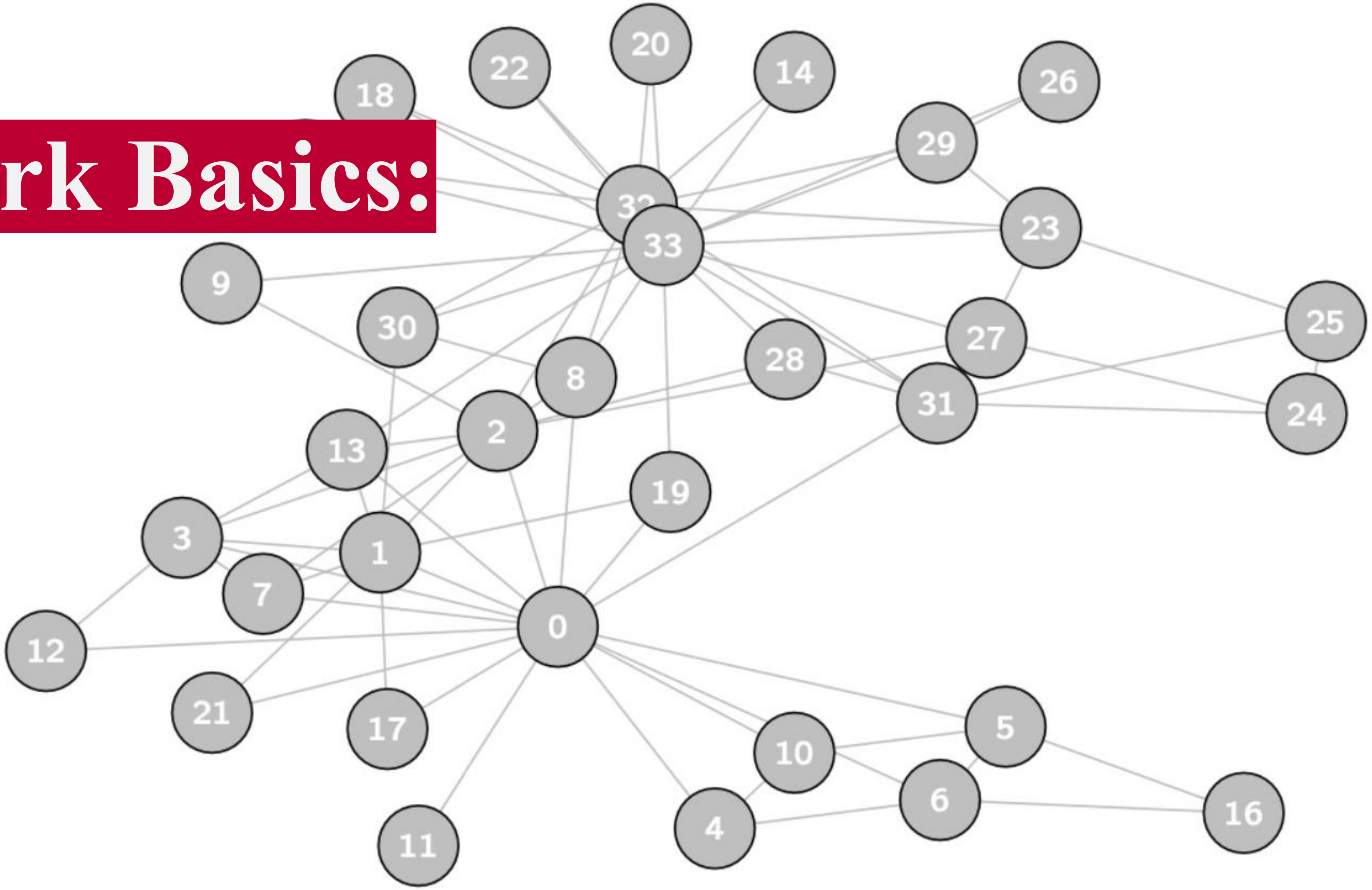
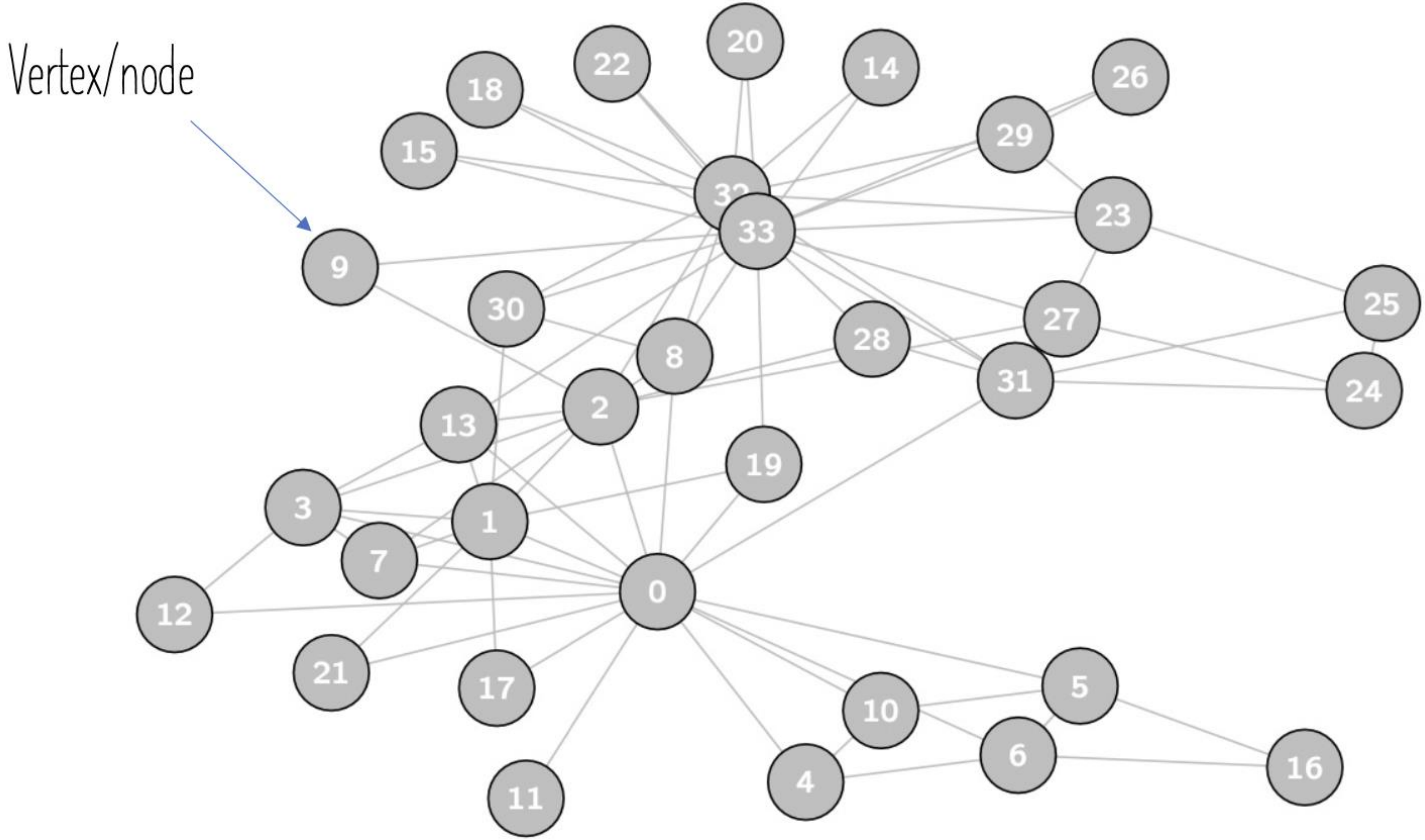
Network Basics:




Vertex/node
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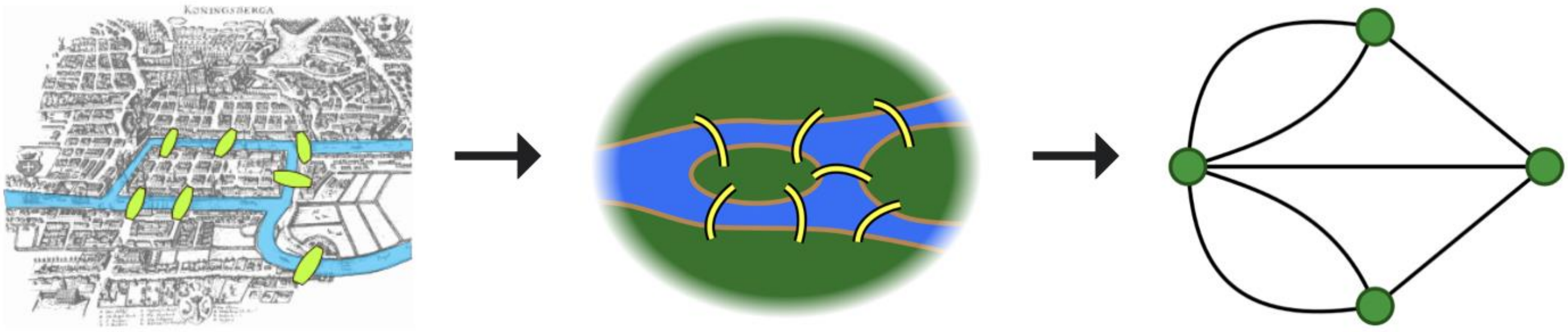
Euler’s approach to the Königsberg Bridge Problem:

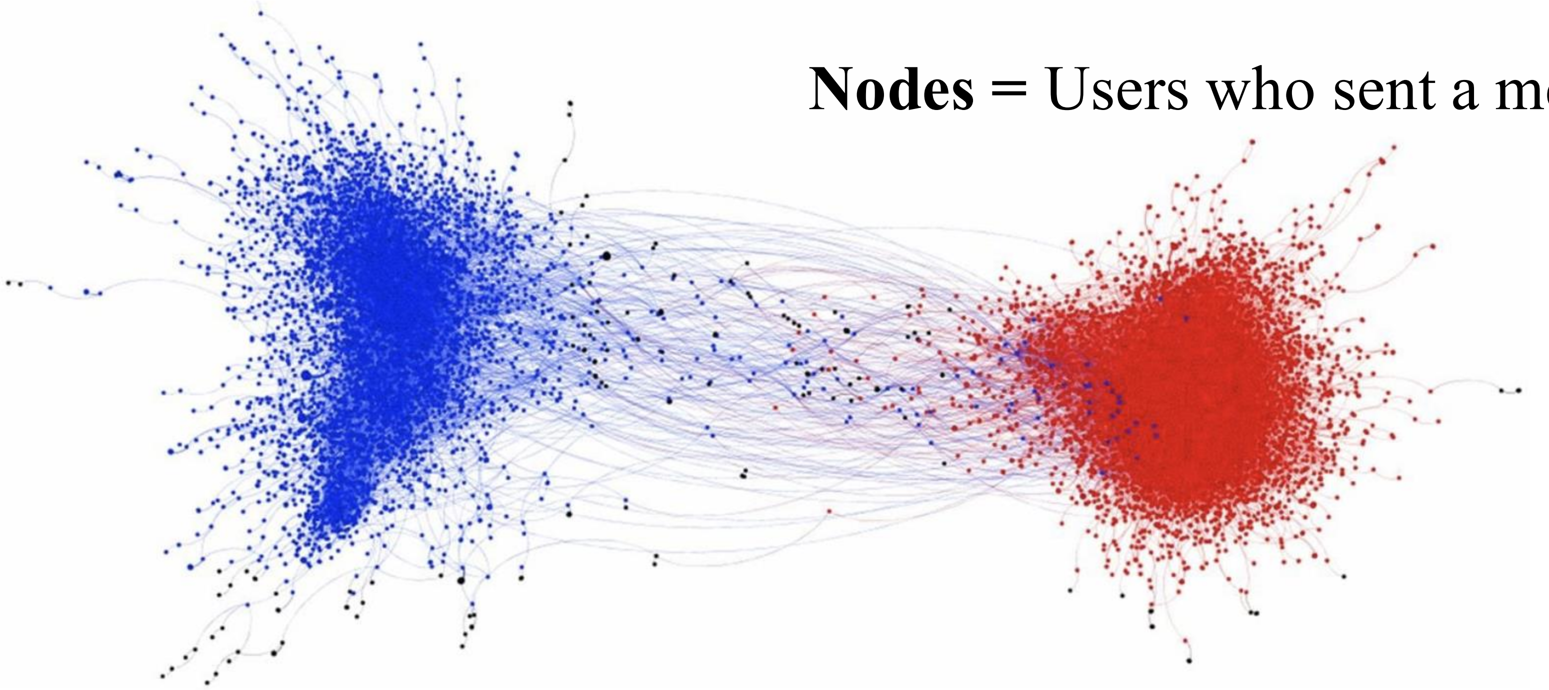
Land mass = Vertex/node

Bridges = edge (which pair of land masses is connected by a bridge?)

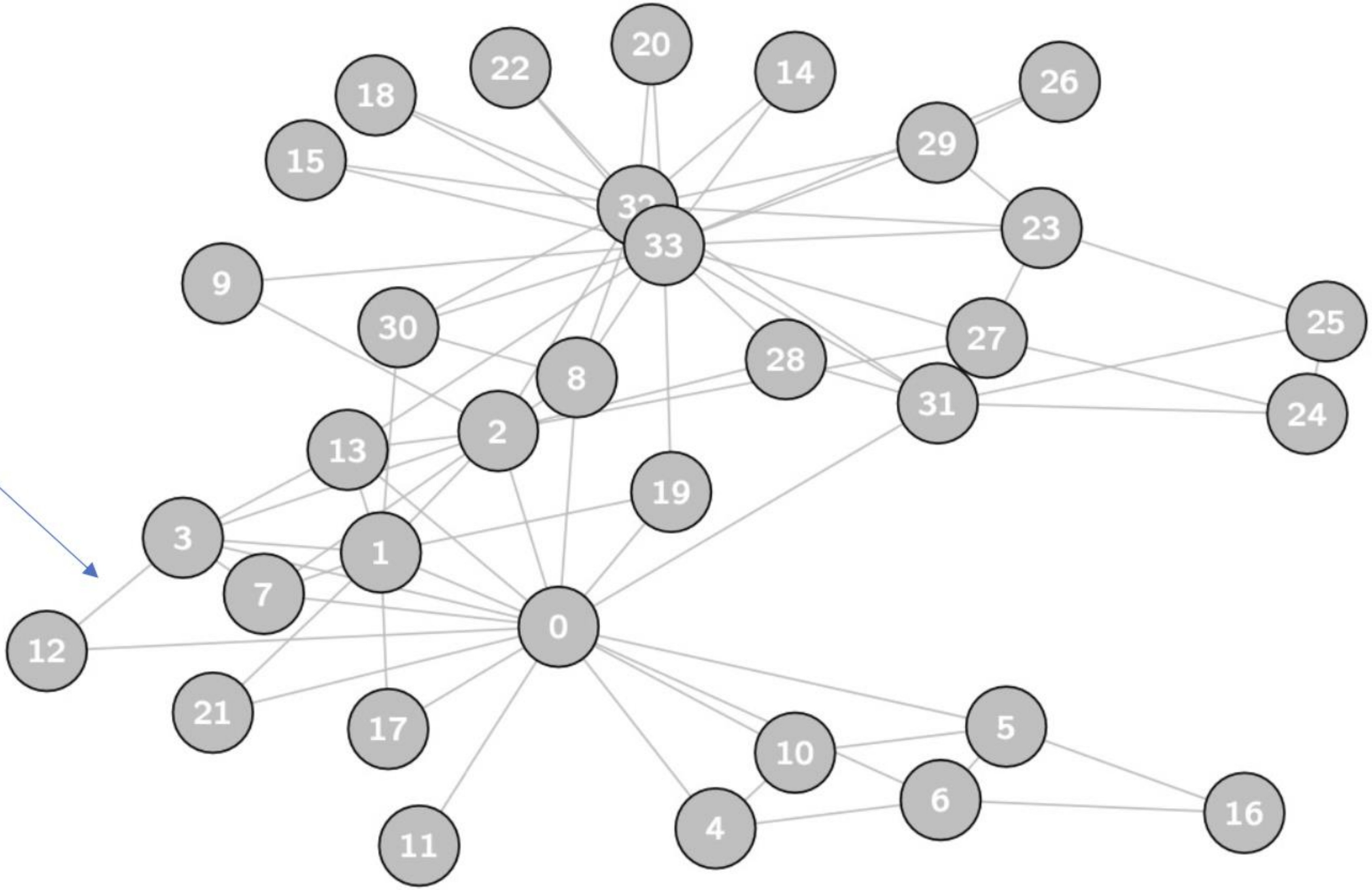


Brady et al., 2017 on Moral Contagion:

Nodes = Users who sent a message.



Network graph of moral contagion shaded by political ideology. The graph represents a depiction of messages containing moral and emotional language, and their retweet activity, across all political topics (gun control, same-sex marriage, climate change). Nodes represent a user who sent a message, and edges (lines) represent a user retweeting another user. The two large communities were shaded based on the mean ideology of each respective community (blue represents a liberal mean, red represents a conservative mean).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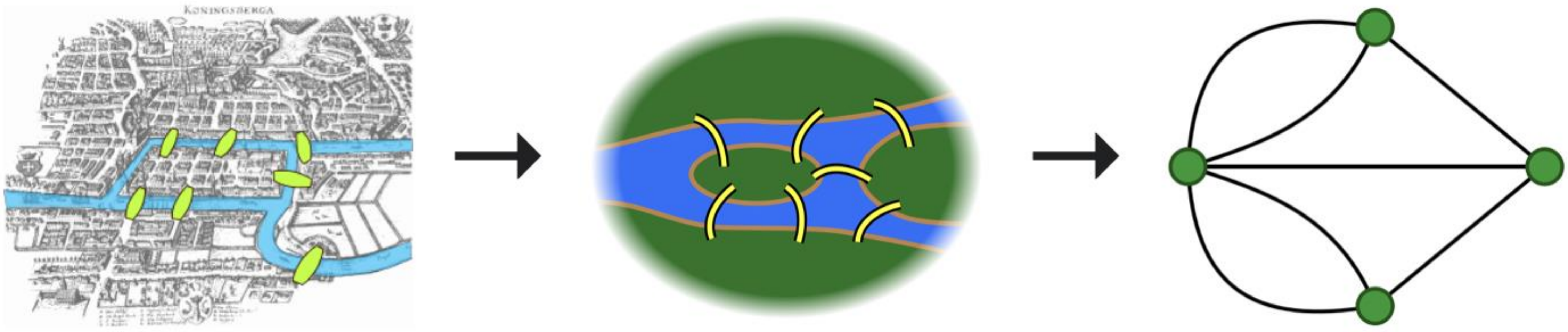
Edge/link

Euler’s approach to the Königsberg Bridge Problem:

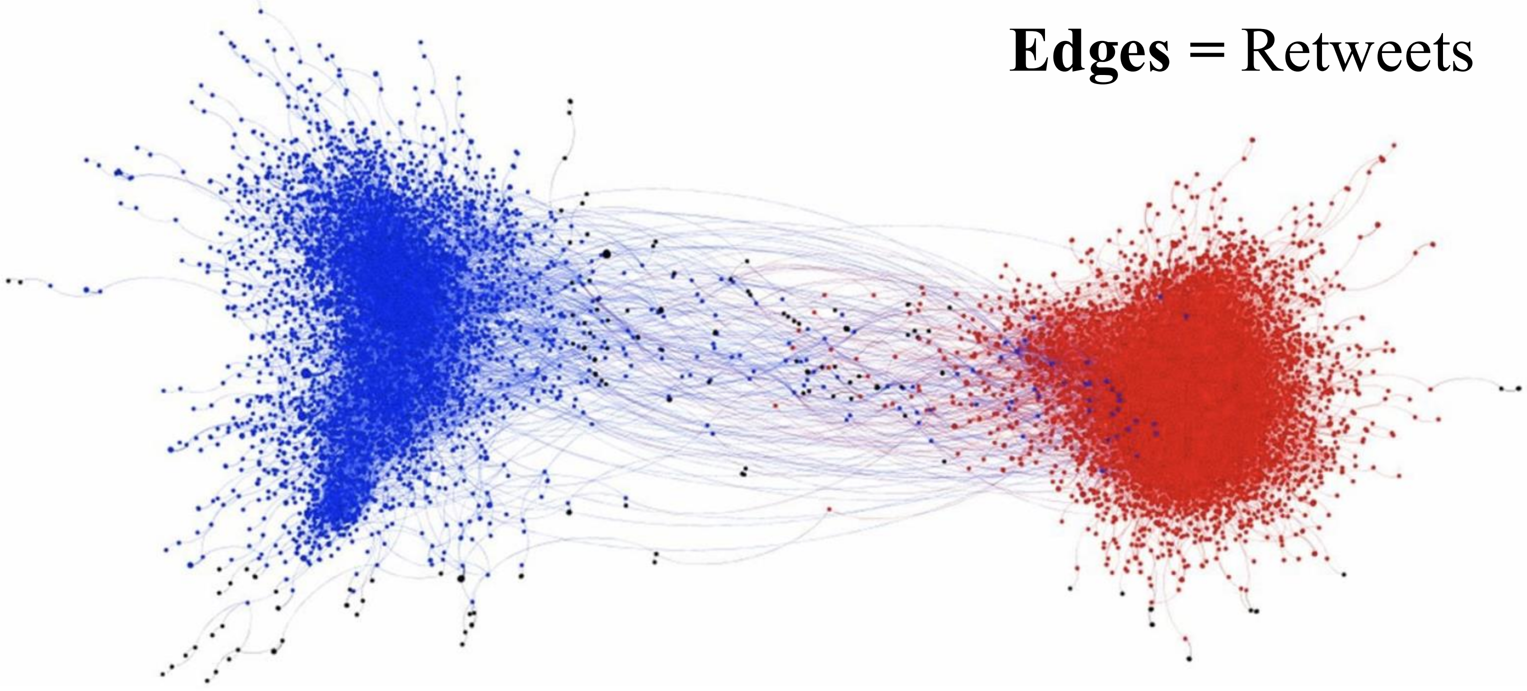
Land mass = Vertex/node

Bridges = edge (which pair of land masses is connected by a bridge?)



Brady et al., 2017 on Moral Contagion:

Edges = Retweets



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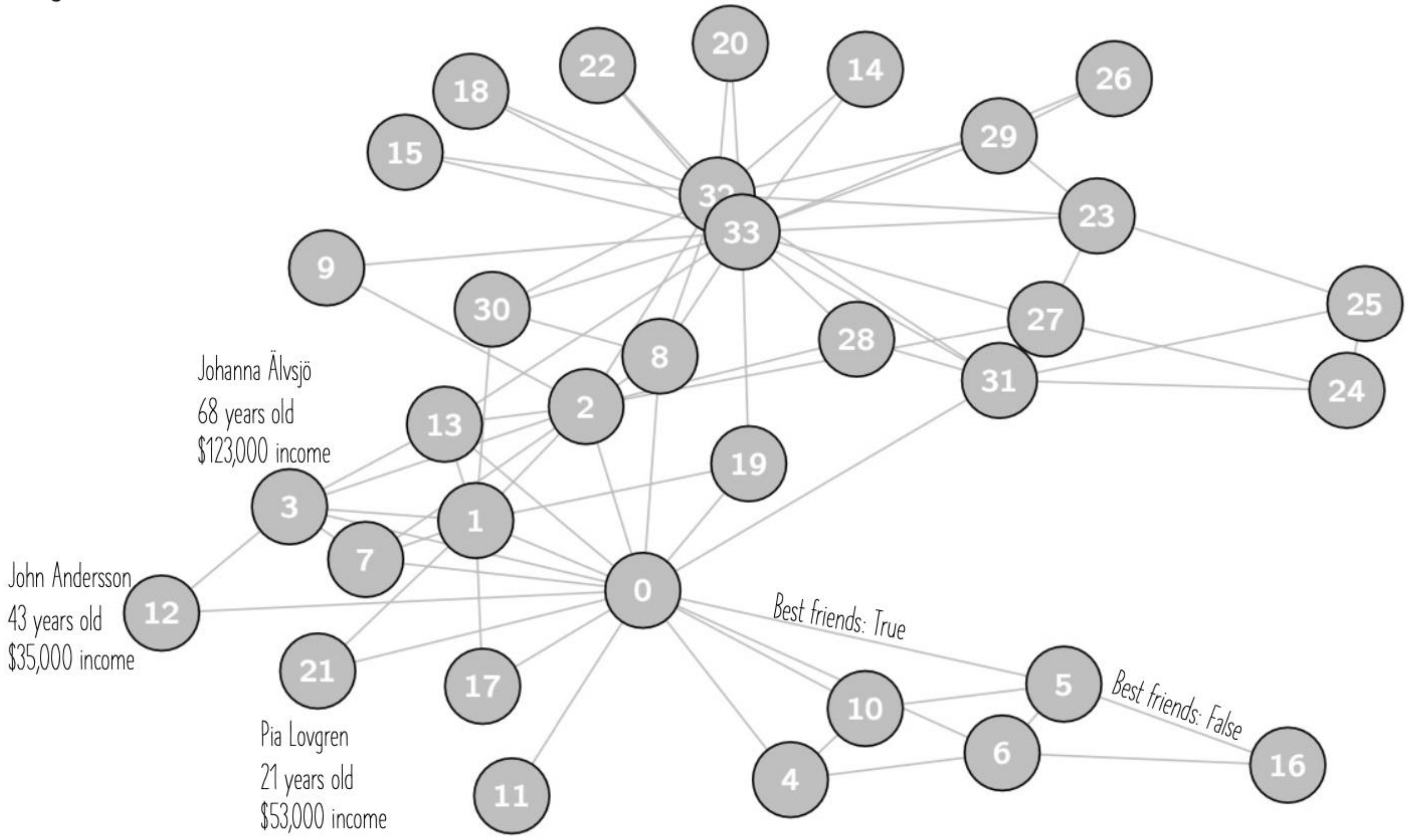
What can nodes and edges represent?

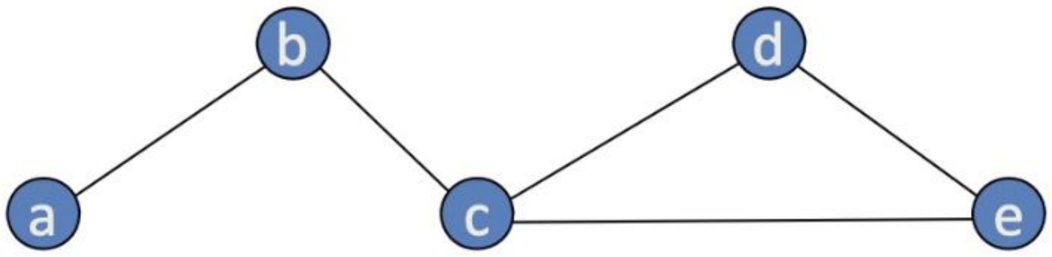
- Basically anything:

Nodes
- People
- Organisations
- Blogs
- Countries

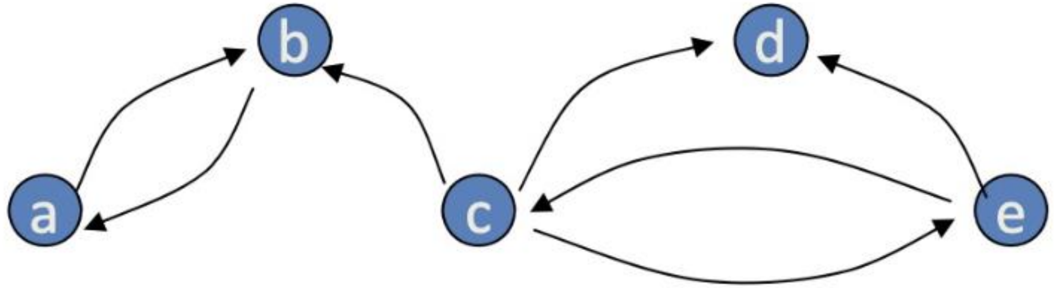
Links
- Communication
- Collaboration
- Hyperlinks
- Alliances

Vertices and edges can have information associated to them

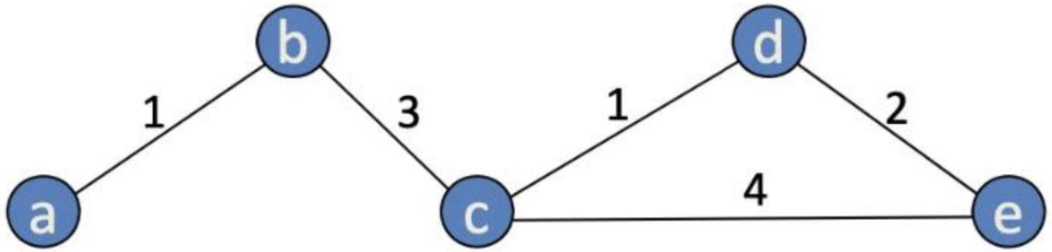




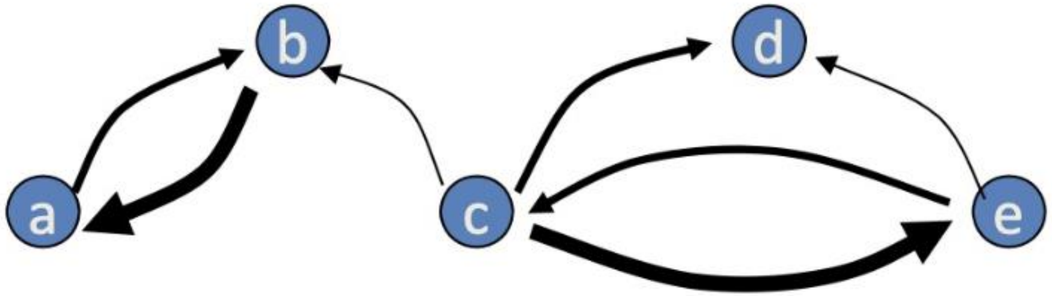
Undirected, binary



Directed, binary



Undirected, Valued



Directed, Valued

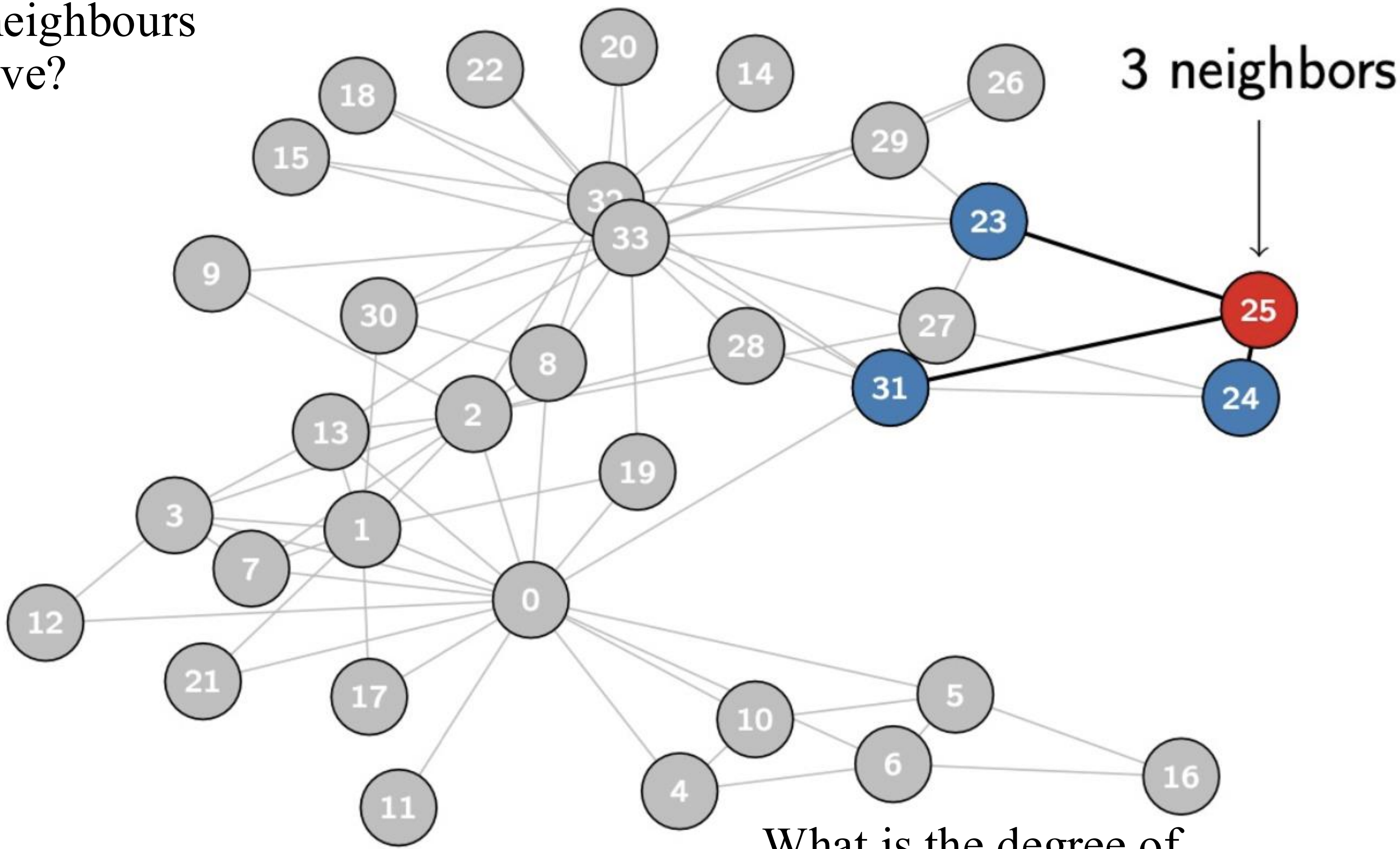
Your project…?

Do you have some network data in your project?
Or can you think of some way of representing it as a network?

- Is it directed?
- Is it weighted?
- What attributes are associated to nodes/edges?

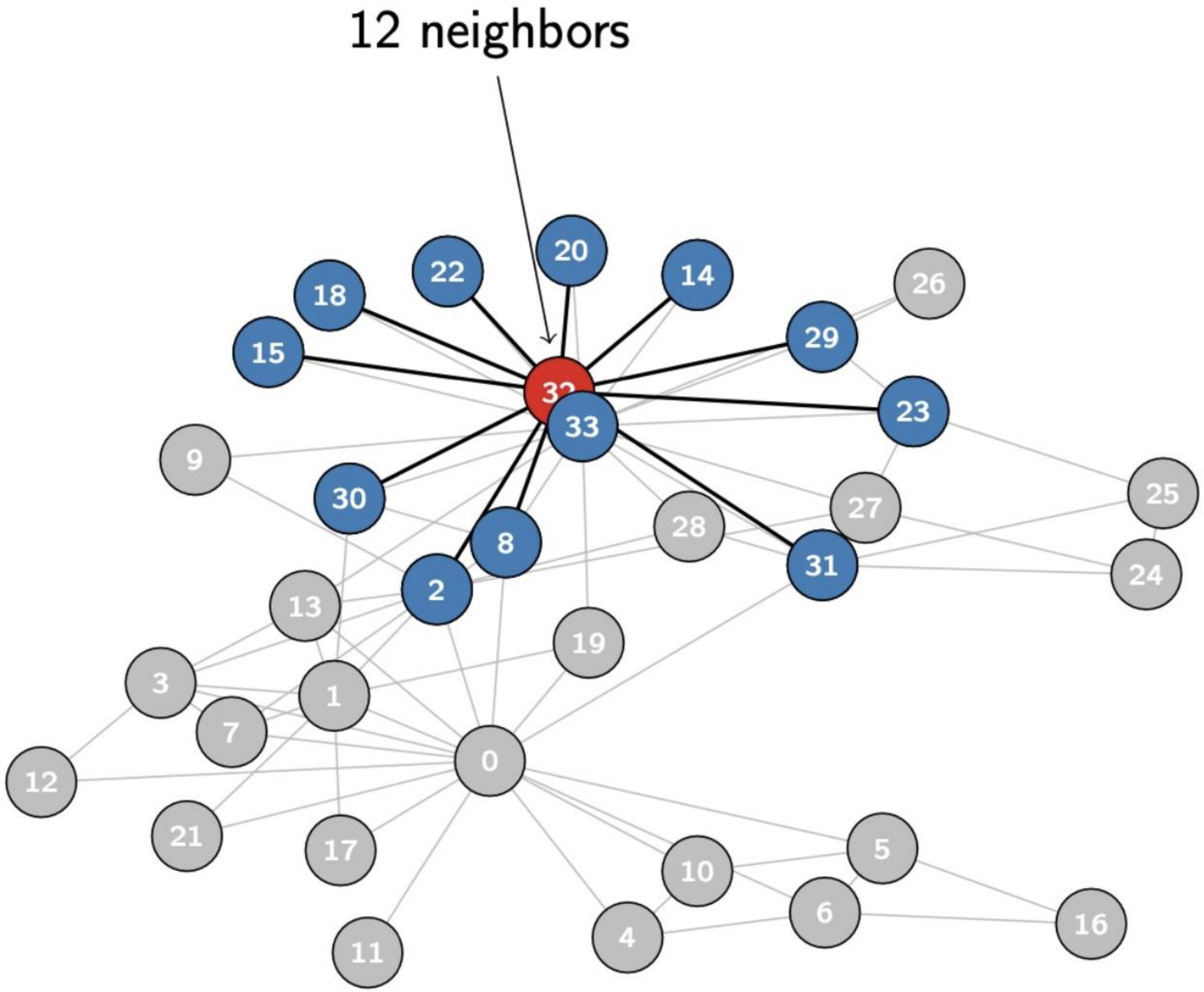
DEGREE

= How many neighbours does a node have?



…What is the degree of node 4?

DEGREE


12 neighbors
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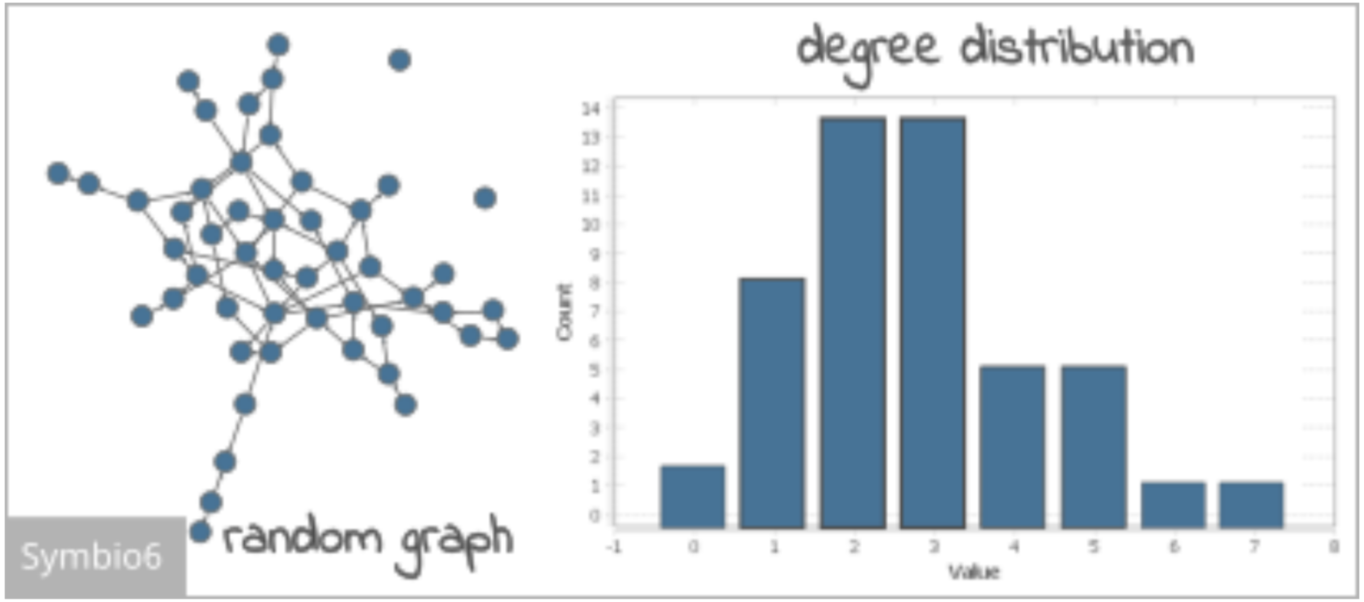
4
11

Degree Distribution

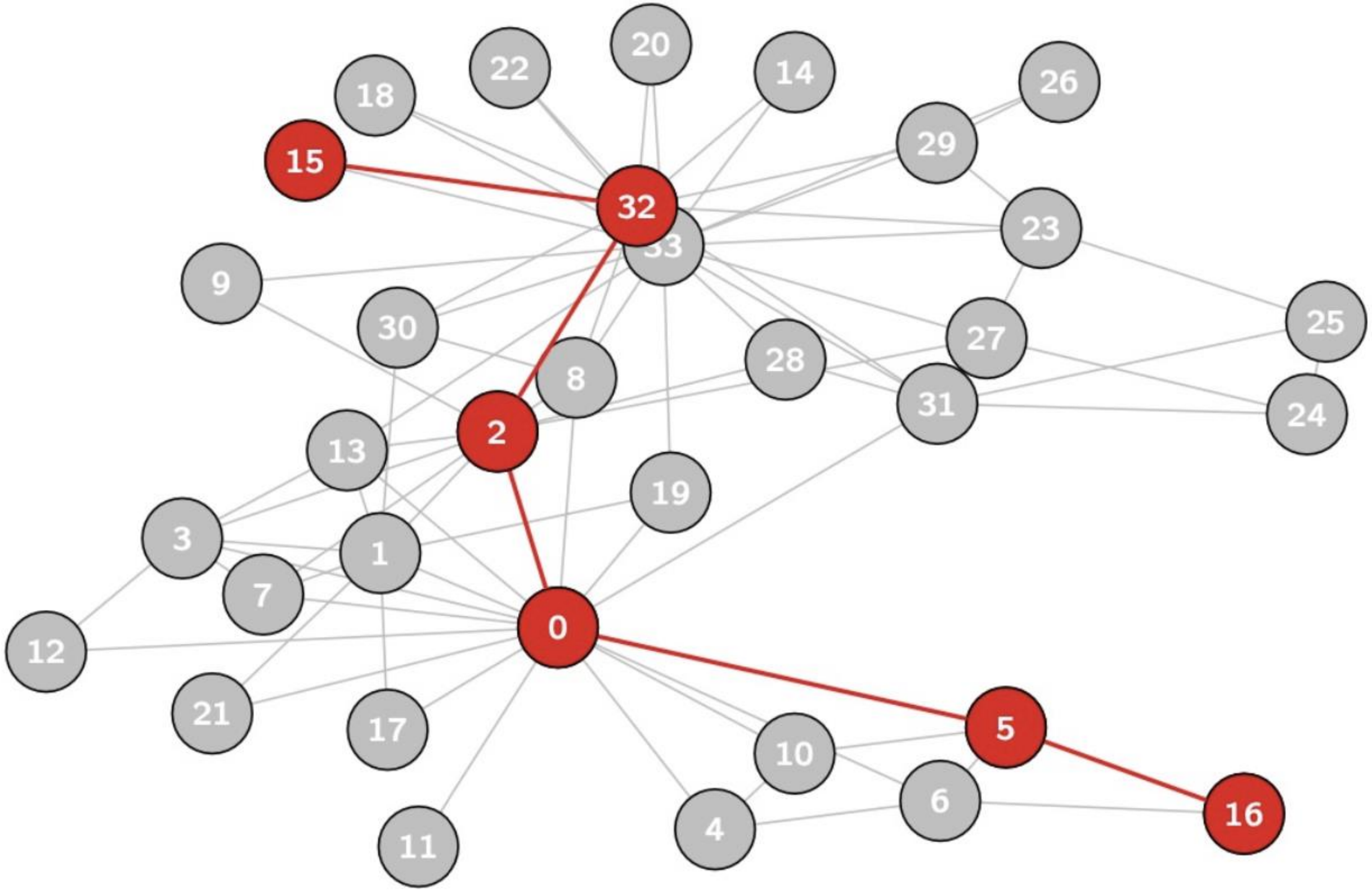
= Number of nodes in a network with each possible degree.

"how many nodes have 1 connection, 2 connections, etc."

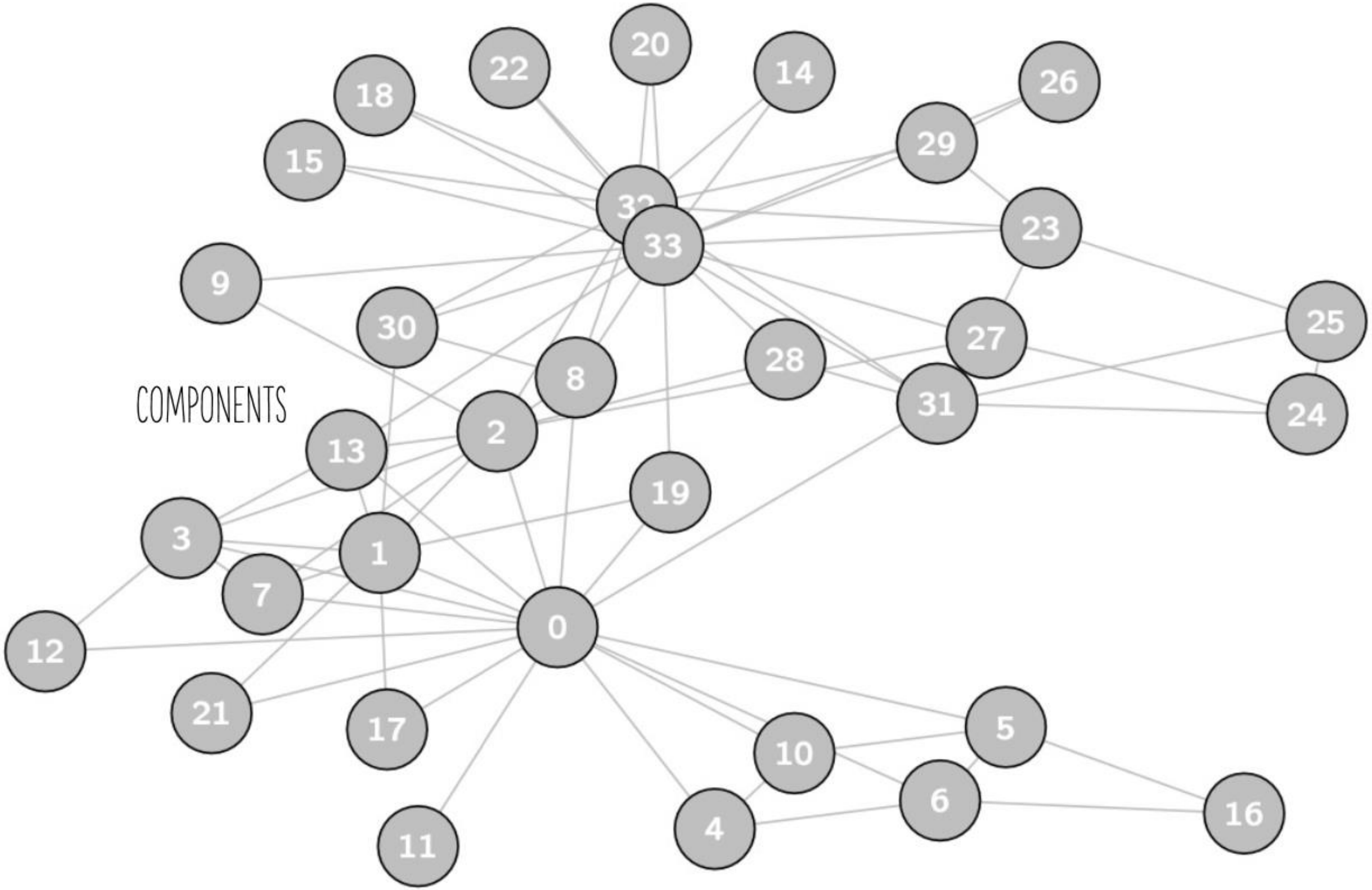


PATHS

= Shortest path between two nodes

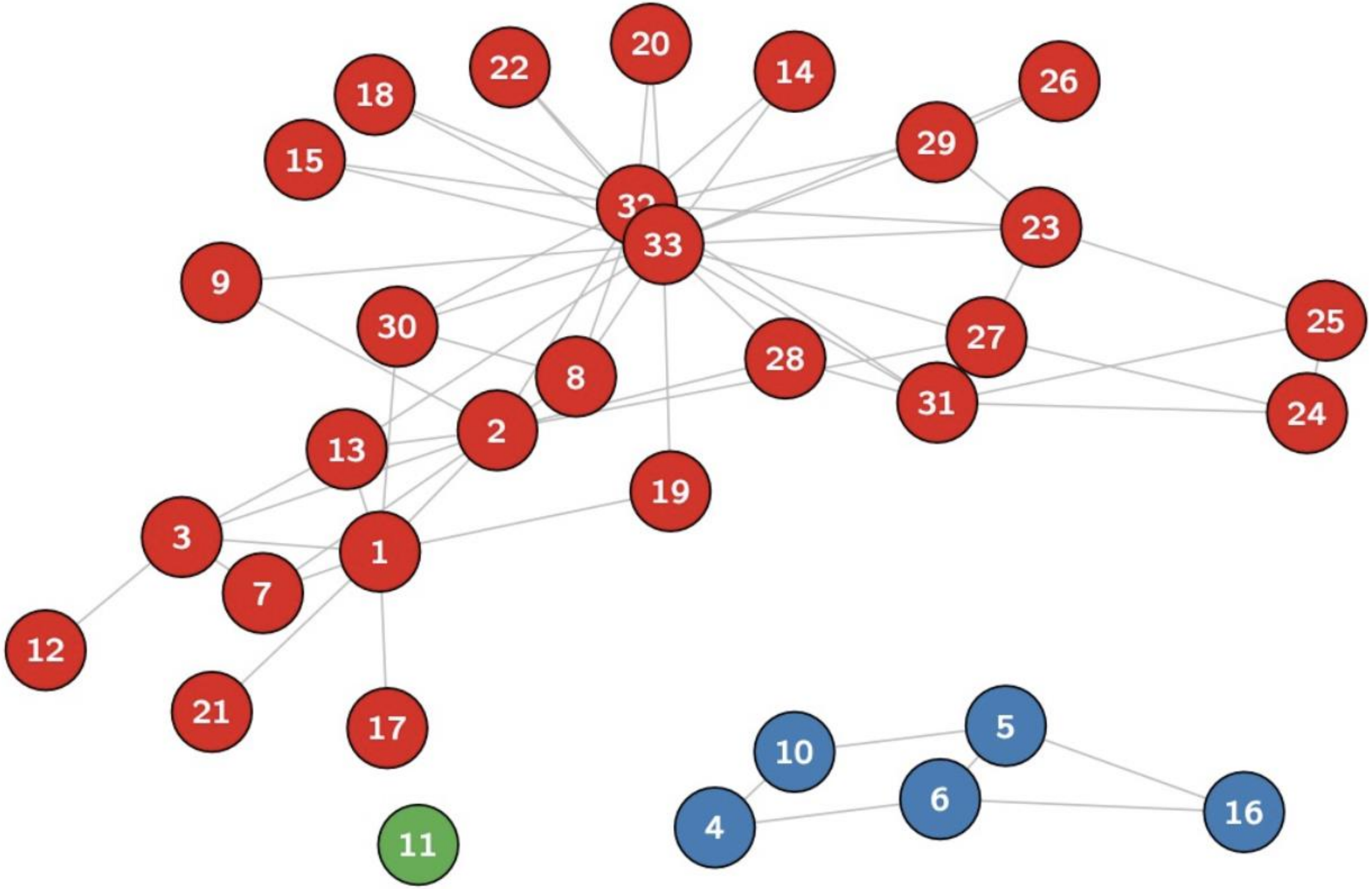


Delete node 0



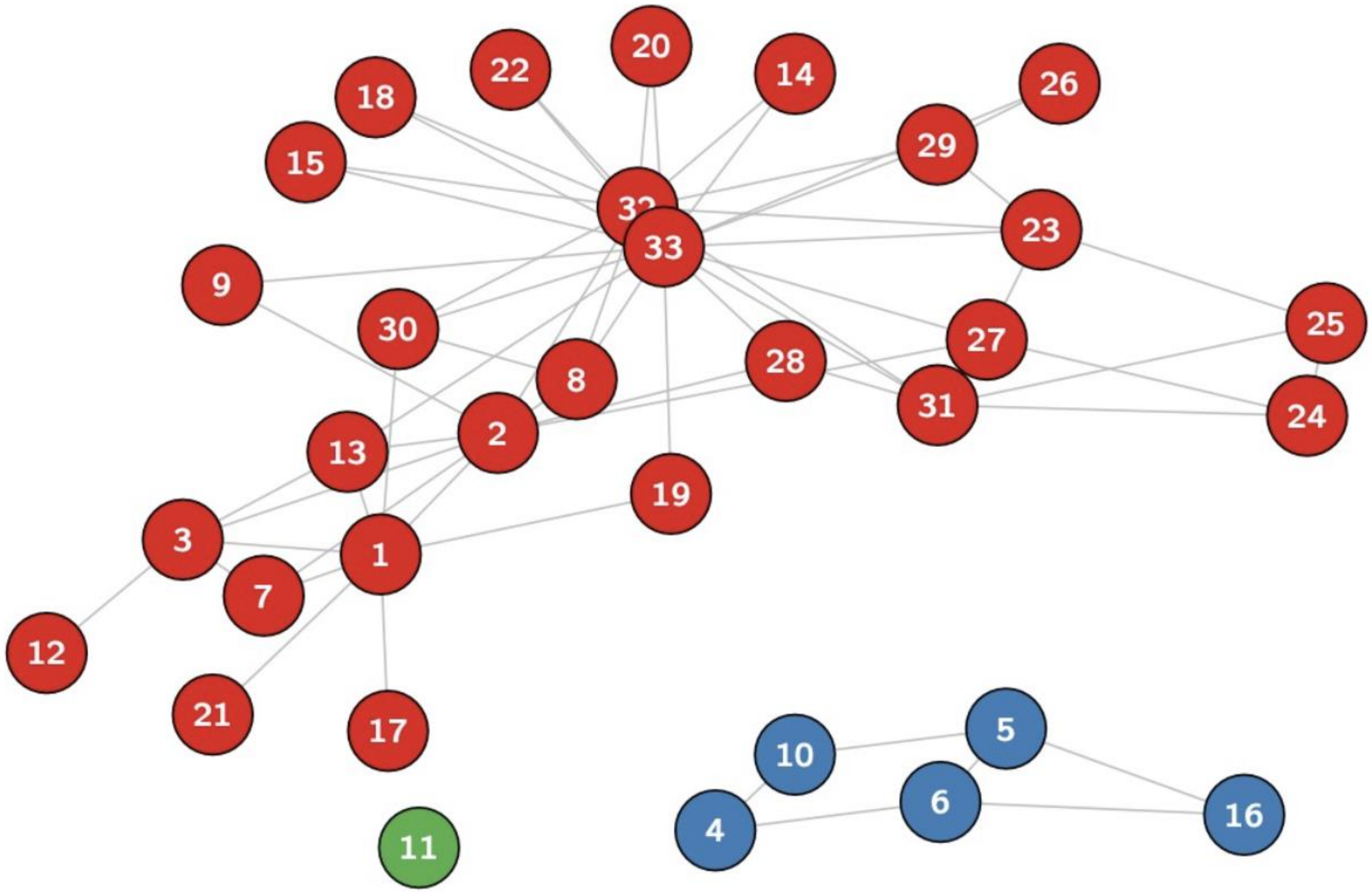
COMPONENTS

Node 0 functioned as a bridge



COMPONENTS

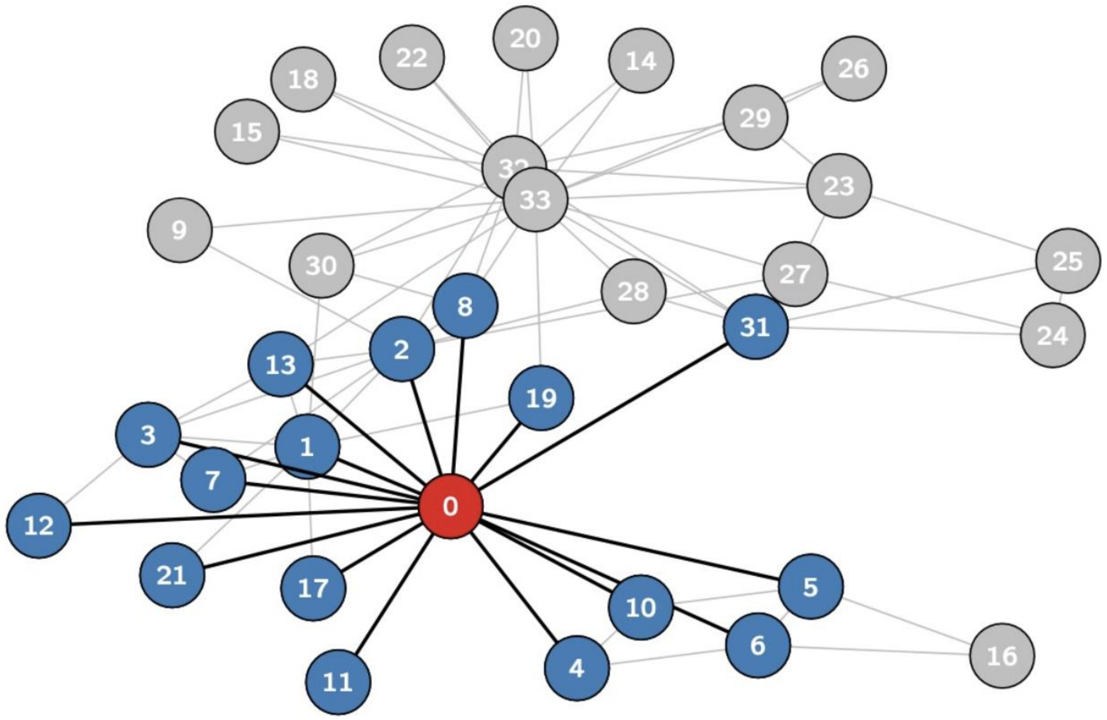
Usually one *giant component*.



Clustering

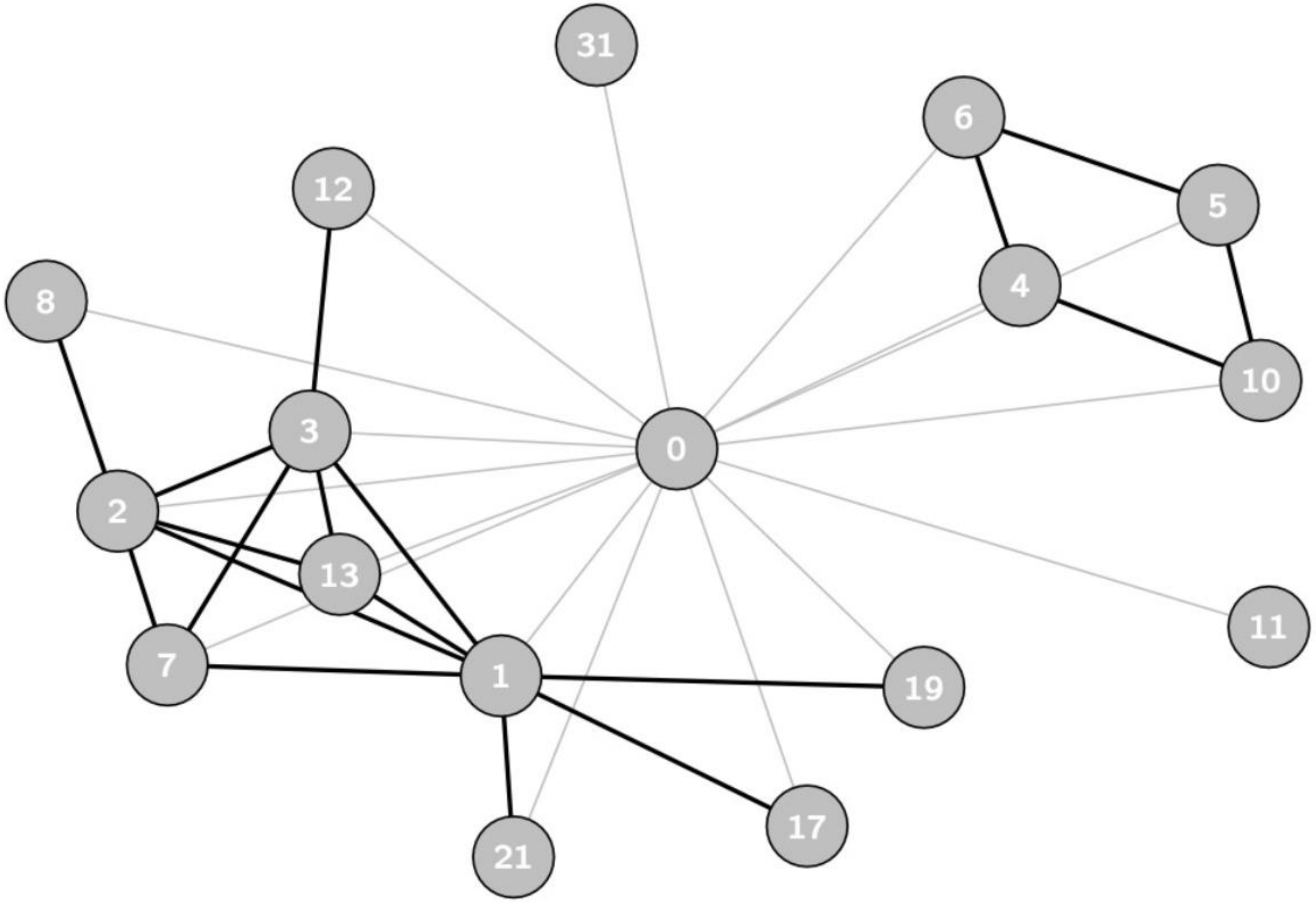
- This measures the degree to which the **neighbors of a specific node** are interconnected.
- **Clustering** refers to the tendency of individuals to form tightly-knit groups characterized by relatively high interconnectivity.

Zoom in on node 0



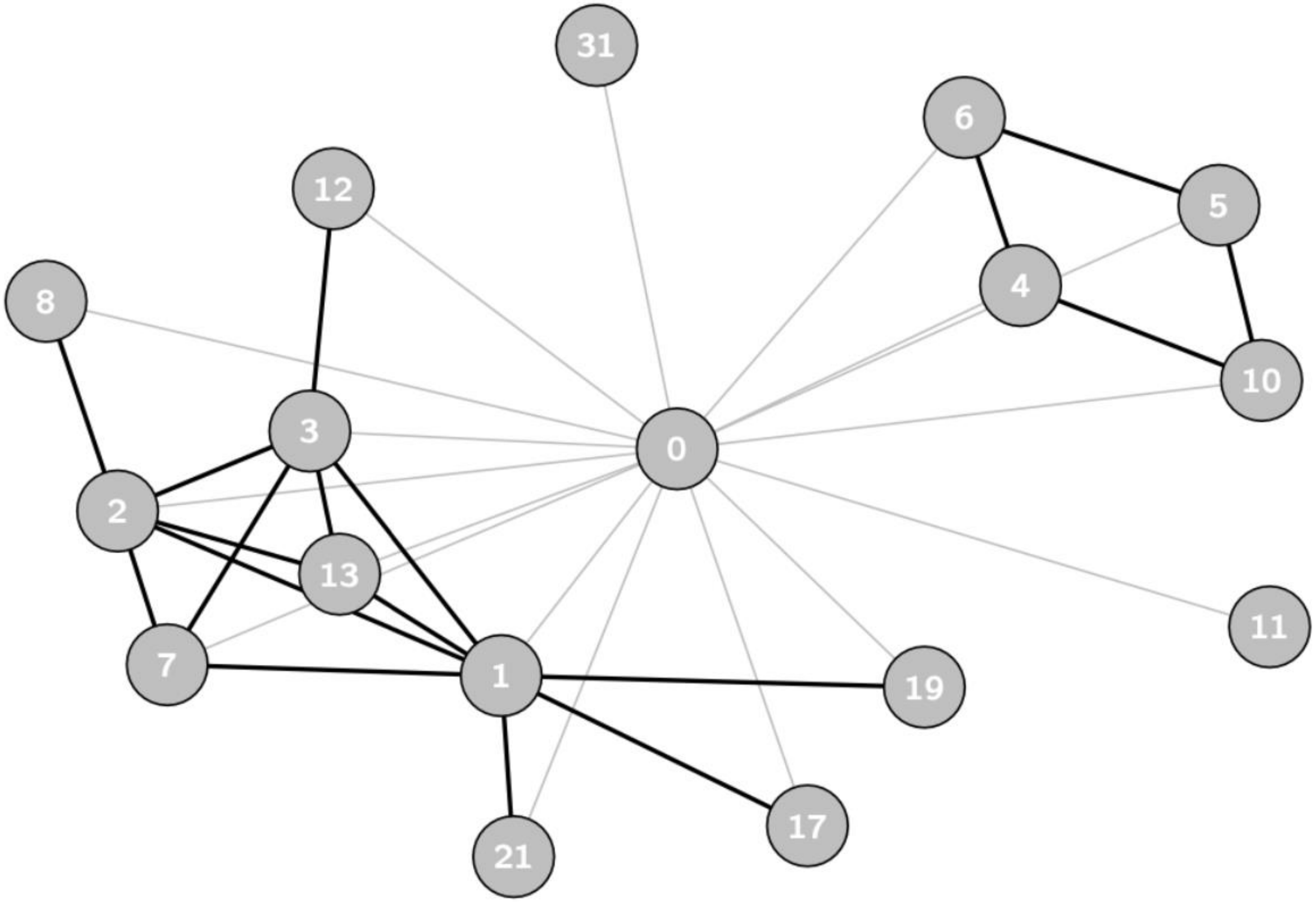
CLUSTERING

Ego network of node 0



CLUSTERING

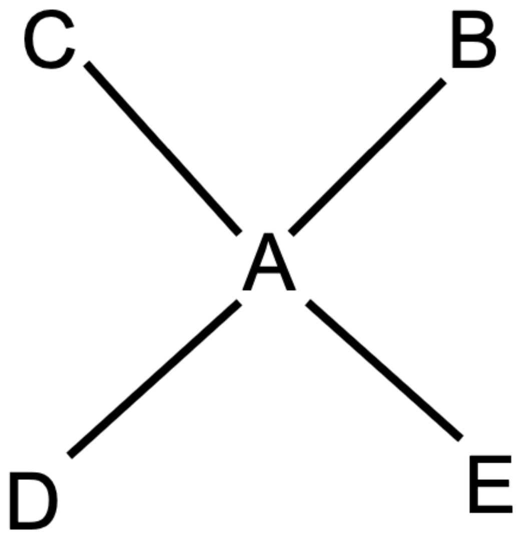
$$\text{Clustering} = \frac{18}{120} = 0.15$$



DENSITY

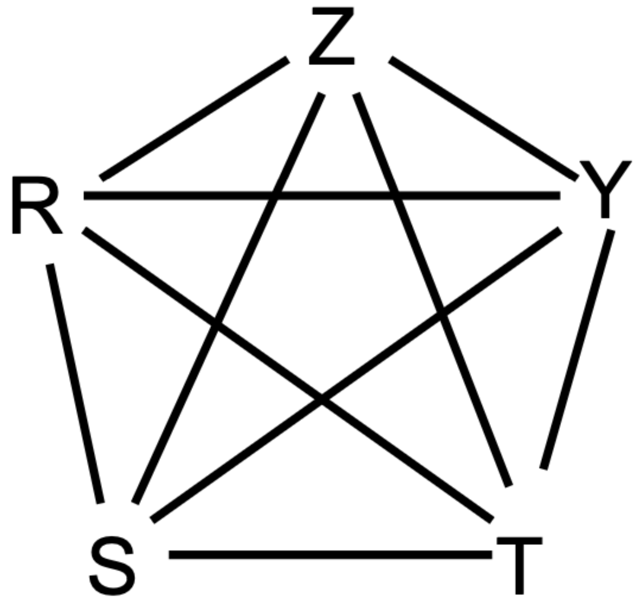
The density of a network is a measure of how many edges are in the network compared to the maximum possible number of edges.

1a



5 nodes, 4 edges

1b



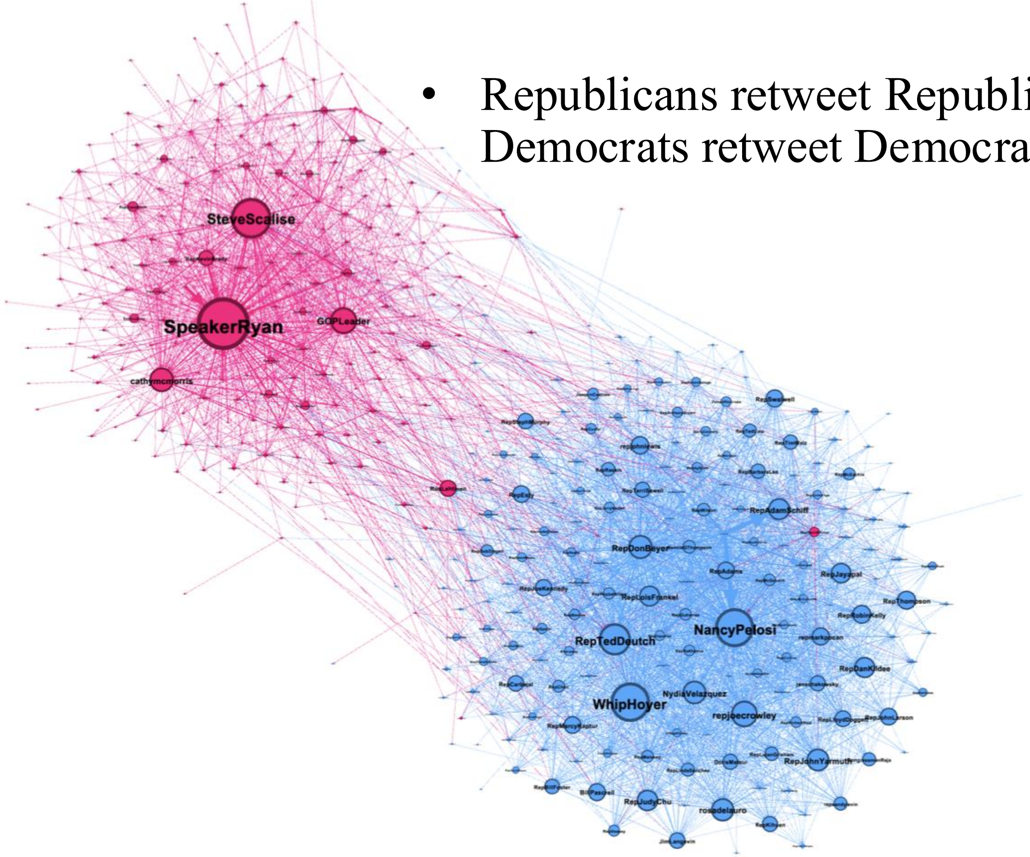
What's the density of this network?

(2*4)/(N(N-1)) = (2*4)/(5(5-1)) = 0.4

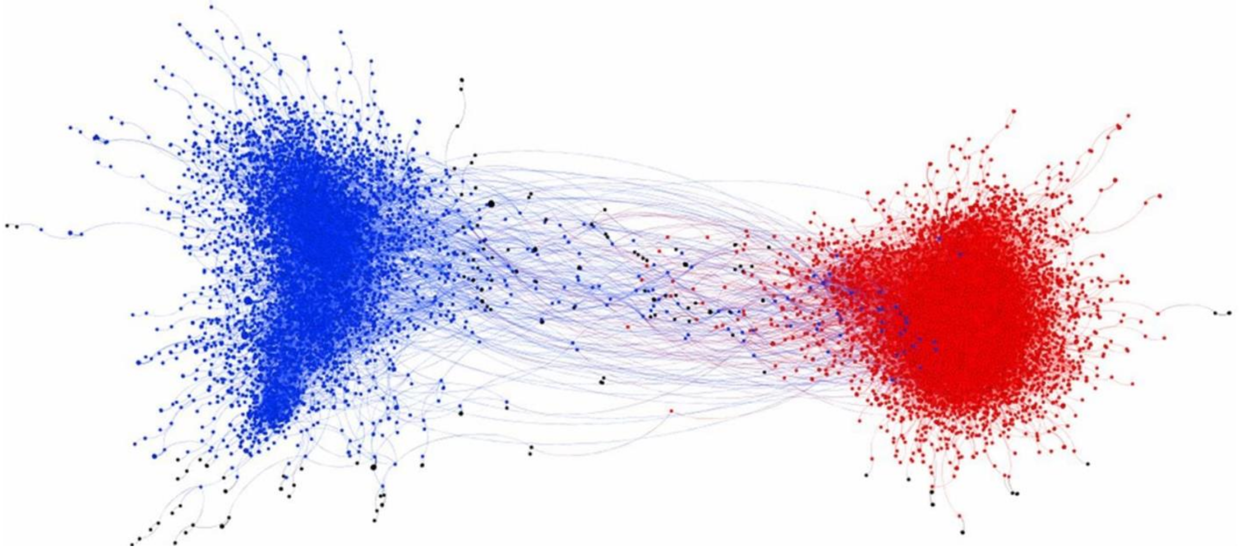
5 nodes, 10 edges

Maximal possible edges: (N * (N-1)) / 2

Homophily: Birds of a Feather Fly Together



- Republicans retweet Republicans, Democrats retweet Democrats



- Idea of "in-group advantage" for *moral contagion* from earlier:
- Liberals tend to retweet moral-emotional language from liberals, conservatives retweet moral-emotional language from conservatives

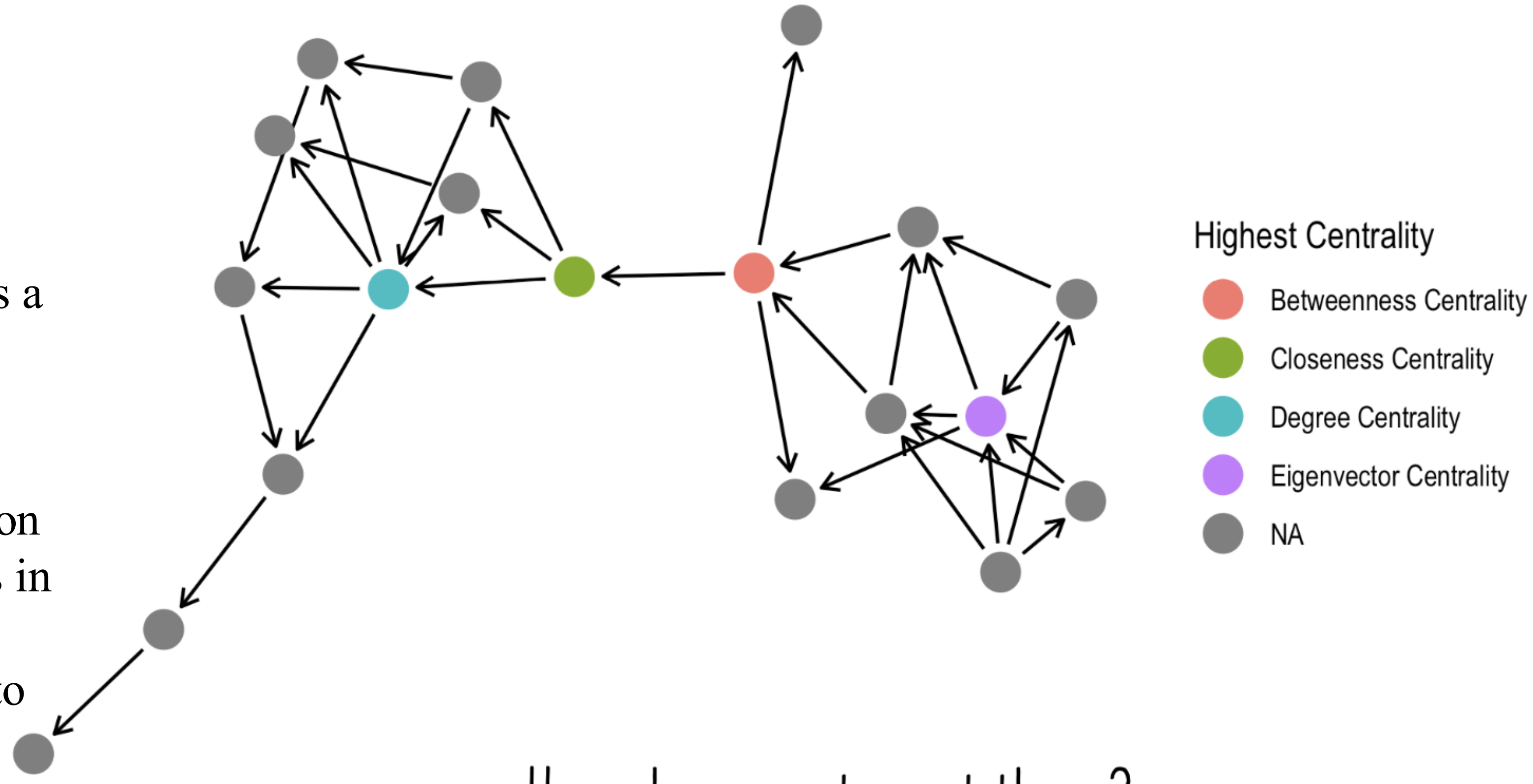
Centrality

Degree centrality: the number of direct connections (or edges) a node has with other nodes in the network.

Betweenness centrality: the number of times a node acts as a bridge along the shortest path between two other nodes.

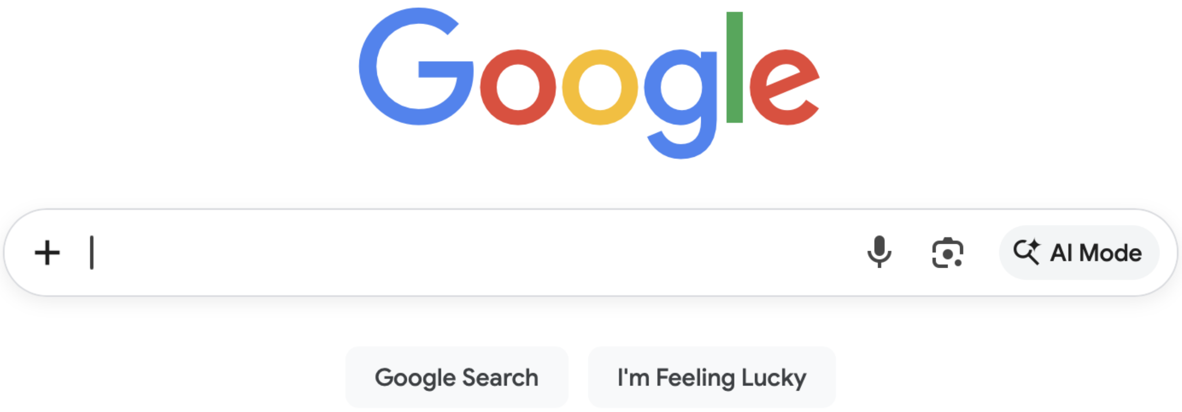
Closeness centrality: how quickly information can flow from a given node to all other nodes in the network, calculated as the inverse of the sum of the shortest distances from that node to all other nodes

Eigenvector centrality: how well- connected your friends are, and how well- connected they are. The probability of encounters in a random walk.



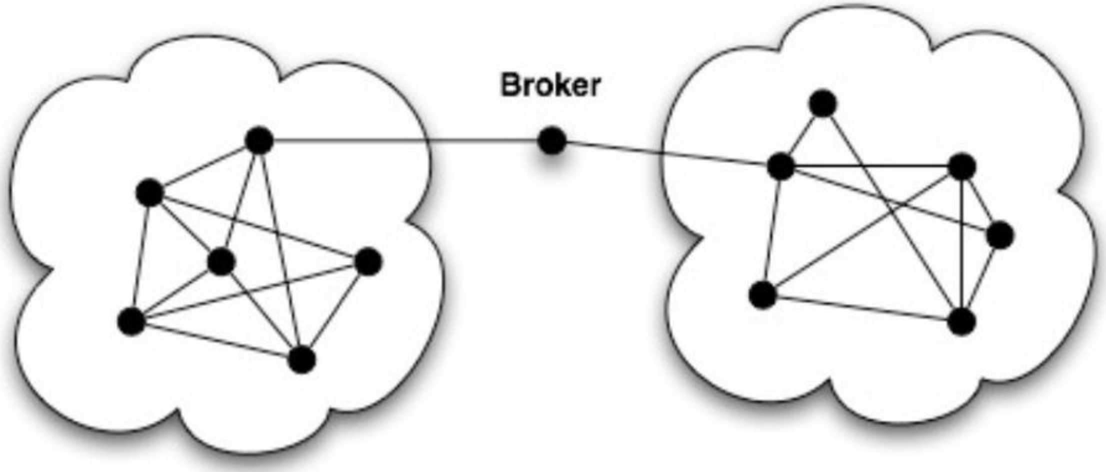
How do you interpret these?

Google Page Rank:



- Google's raking of search results
- One of the mechanisms: Eigenvector centrality
- → How many incoming links from already important pages does a page have?

BROKERAGE

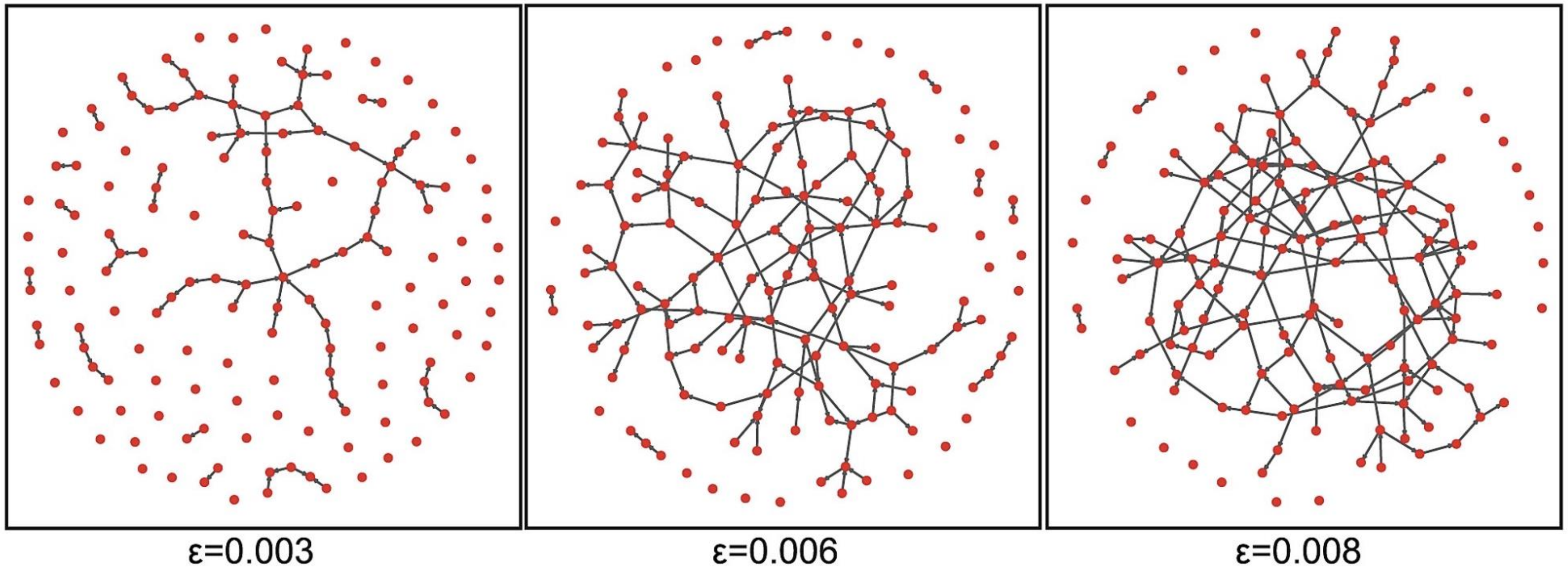


Brokers can be powerful,
as they control the flow of information

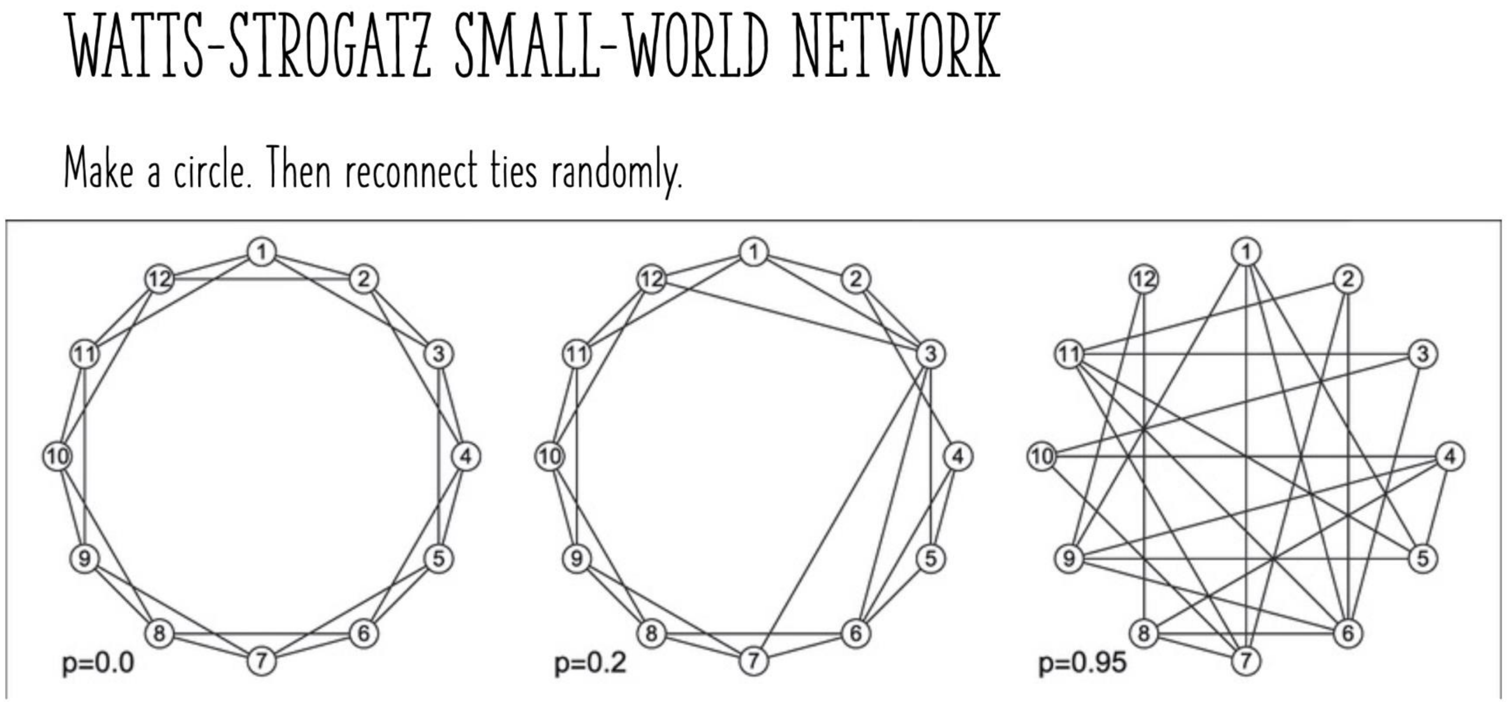
Generating Networks

ERDÓS-RÉNYI GRAPH (RANDOM GRAPH):

Nodes are connected randomly with a fixed probability.



Generating Networks



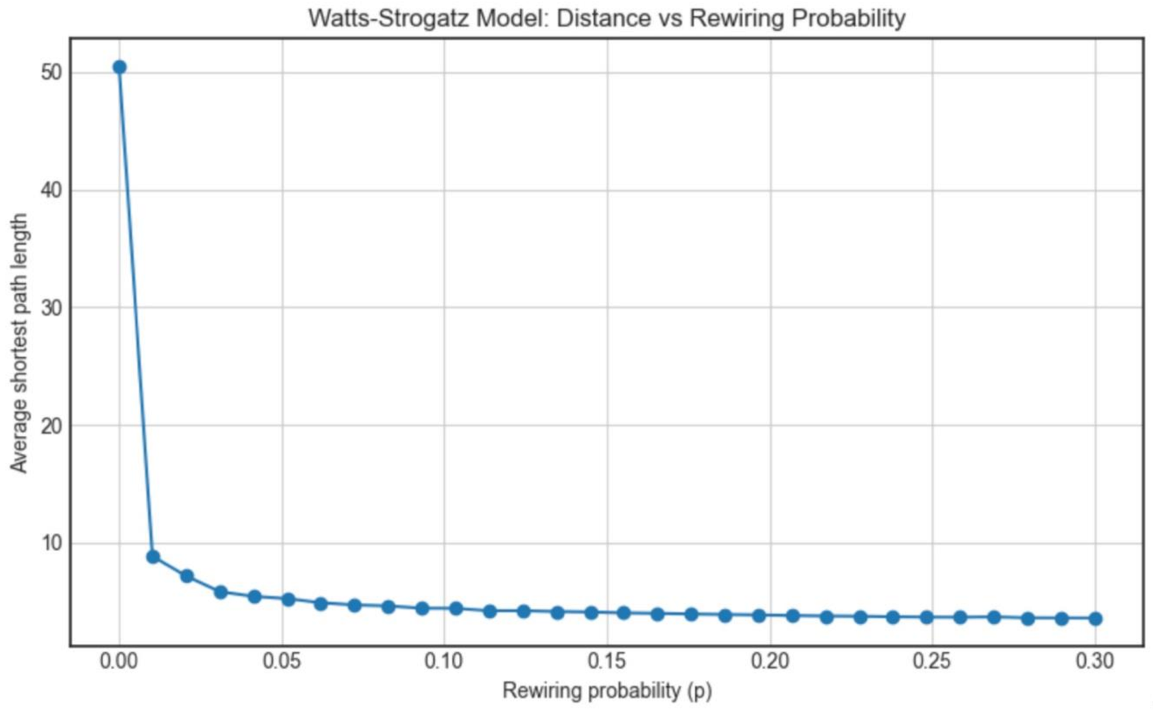
Graphs with **small-world properties:**

- Short average path lengths
- High clustering

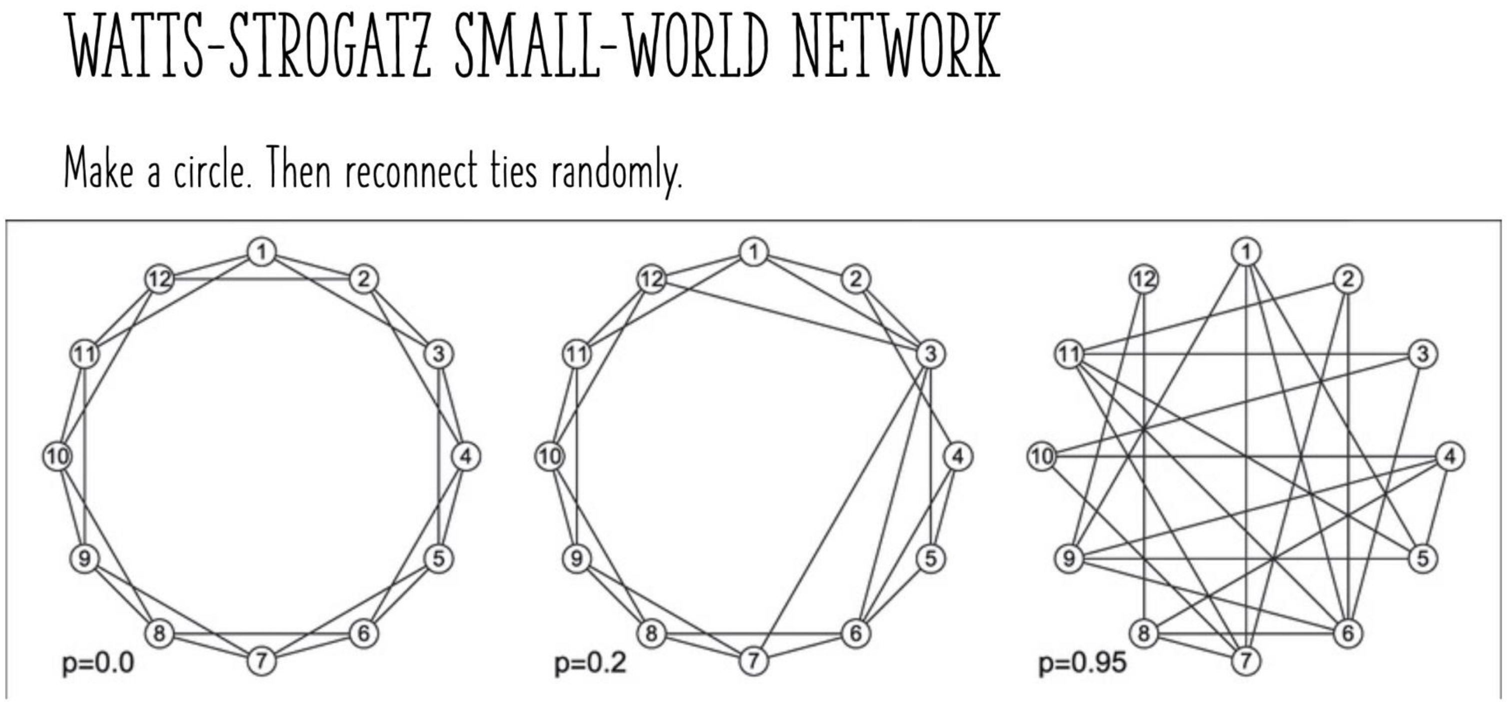
Generating Networks

HOW DOES THE AVERAGE SHORTEST PATH CHANGE WITH REWIRING?

What does this mean?



Generating Networks



Graphs with **small-world properties:**

→ **Social networks:**

High Clustering = high probability that two friends of one person are friends themselves

Short average path lengths = A surprisingly short chain of connections between any two people

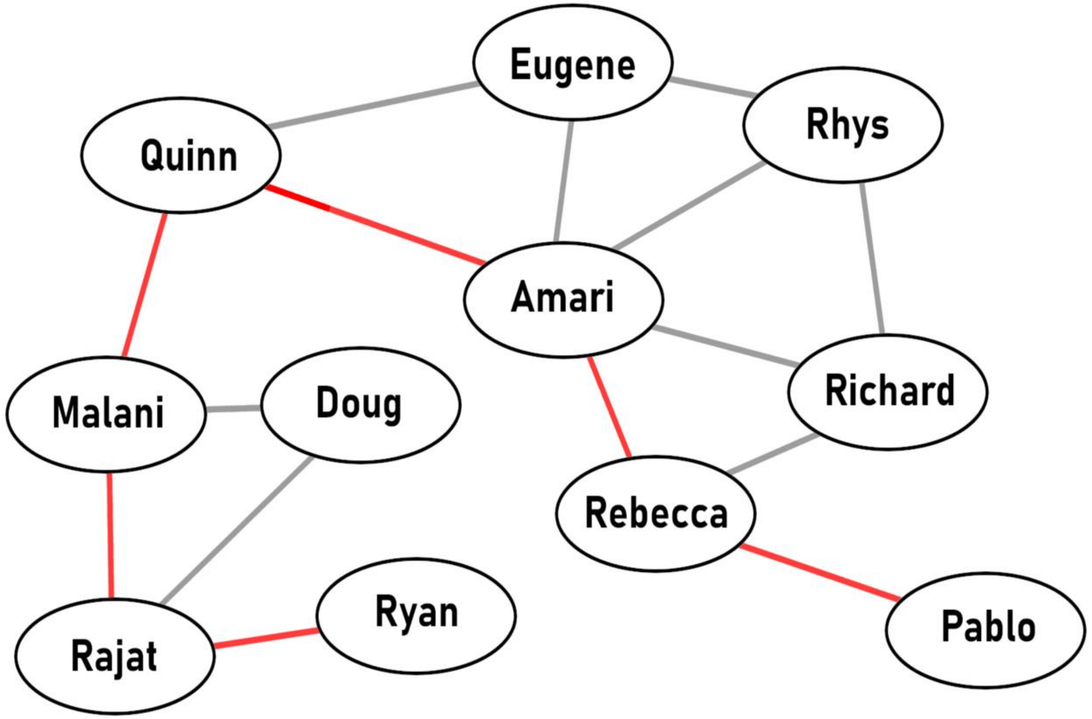
Small world experiment

- Stanley Milgram
- Picked 300 random people in Omaha
- Asked them to get a letter to a stockholder in Boston by passing it through friends

- How many steps did this take?
- Letters arrived on average: 6.2 steps
- Total of 64 chains

Six Degrees of Separation

- Idea: People are separated by six or less social connections
- By passing through a chain of “friends of friends”, a max. of 6 steps are needed to connect any two people in the network



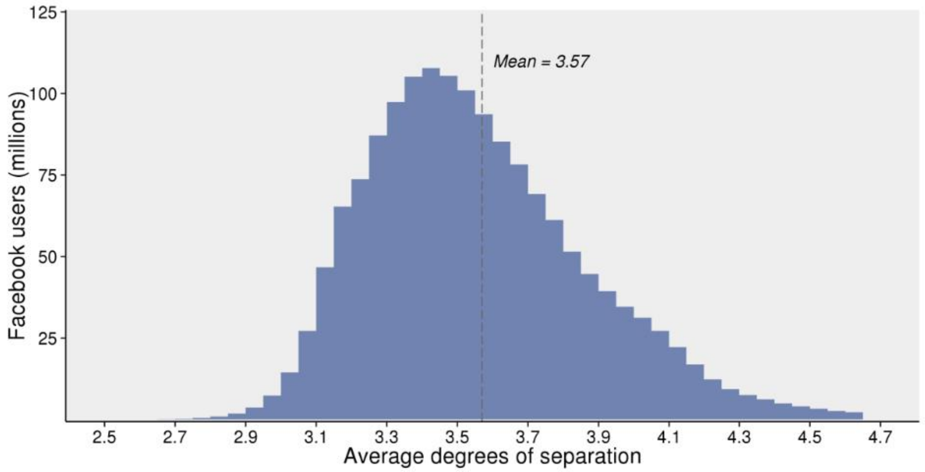
Six Degrees of Separation

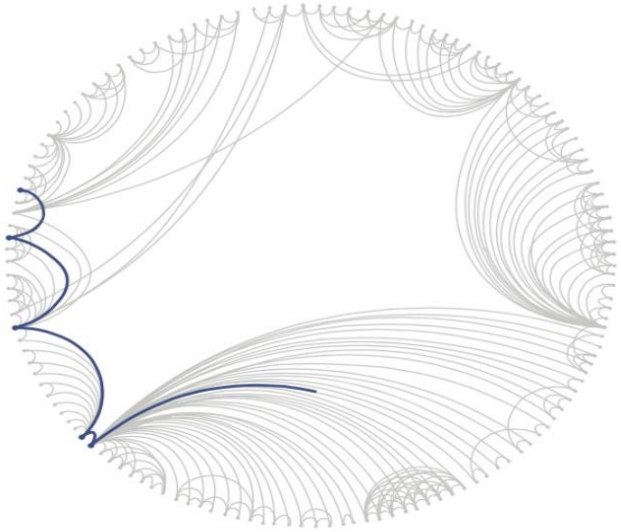
1. You have 100 friends	100
2. Your friends have 100 friends	100*100 = 10,000
3. Your friends' friends have 100 friends	100*100 = 10,000
4. Your friends' friends' friends have 100 friends	100*100*100 = 1,000,000
5. Your friends' friends' friends' friends have 100 friends	100*100*100*100 = 100,000,000
6. Your friends' friends' friends' friends' friends have 100 friends	100*100*100*100*100 = 10,000,000,000
....	
n. Your friends' ... friends have 100 friends	100^n


acebook
December

Path length on Facebook

- Each person in the world (at least among the 1.59B people actively using Facebook) is connected to every other person by an **average of 3.5 other users.**
- Average distance we observe = 4.57
- This corresponds to 3.57 intermediaries or "degrees of separation"
- Within the United States = 3.46 degrees of separation.

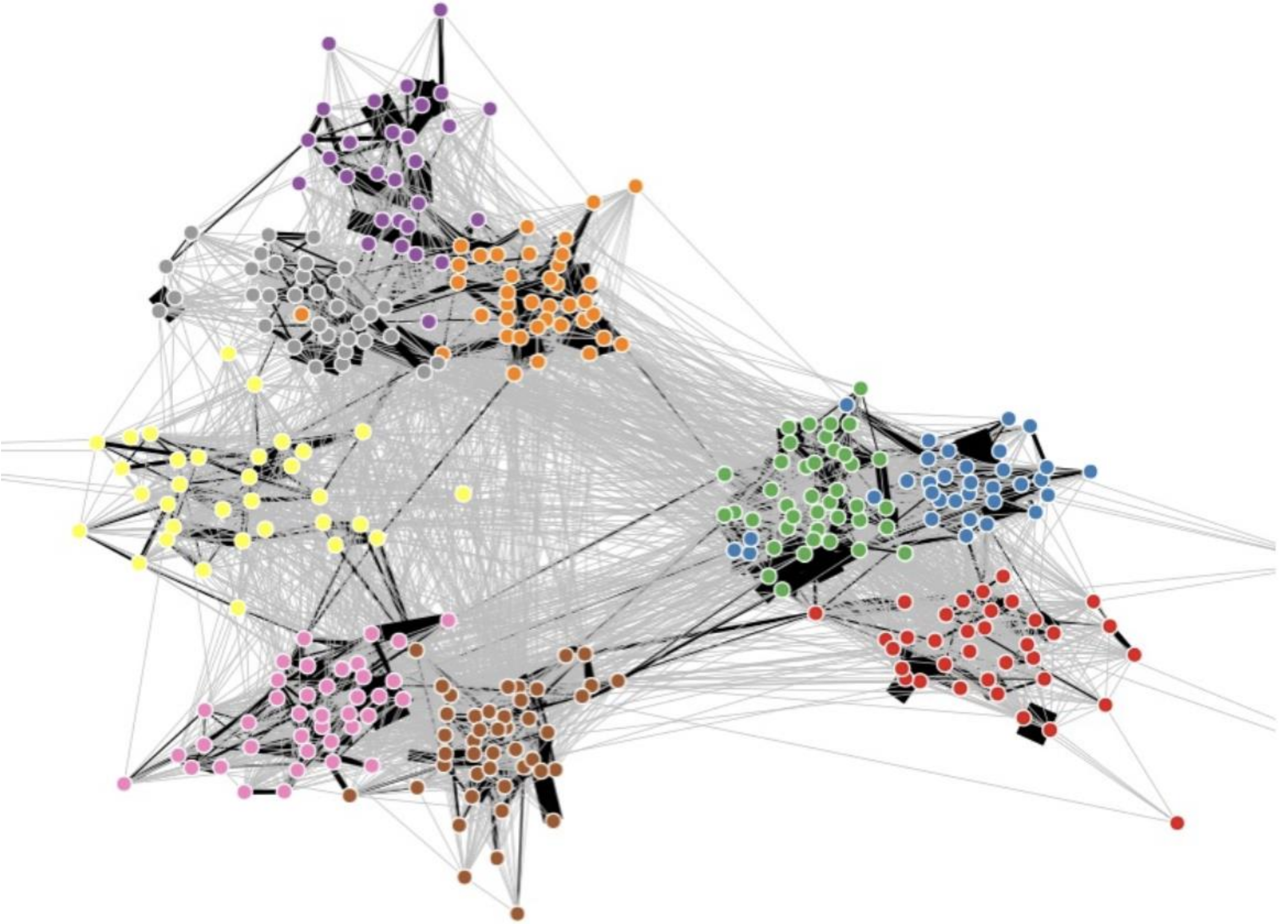




The Strength of Weak Ties

- Weak ties are links that have a low weight
- Strong ties are our close friends. These tend to be clustered.

- **Mark Granovetter:** weak social ties (e.g., acquaintances), are often more valuable than strong ties (e.g., close friends) for accesssing new information, opportunities or resources. Weak ties form bridges to different networks, providing access to diverse, non-redundant information that isn't available within one's close-knit social circle.



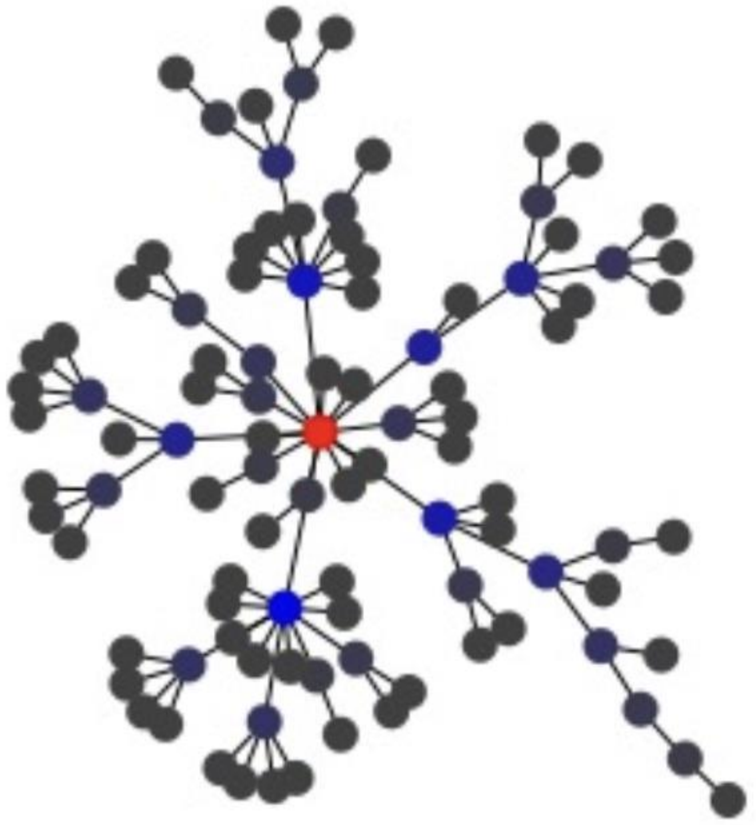
Barabási–Albert – preferential attachment

- Each new node connects to existing nodes with a probability proportional to how many links they already have: "the rich get richer.”

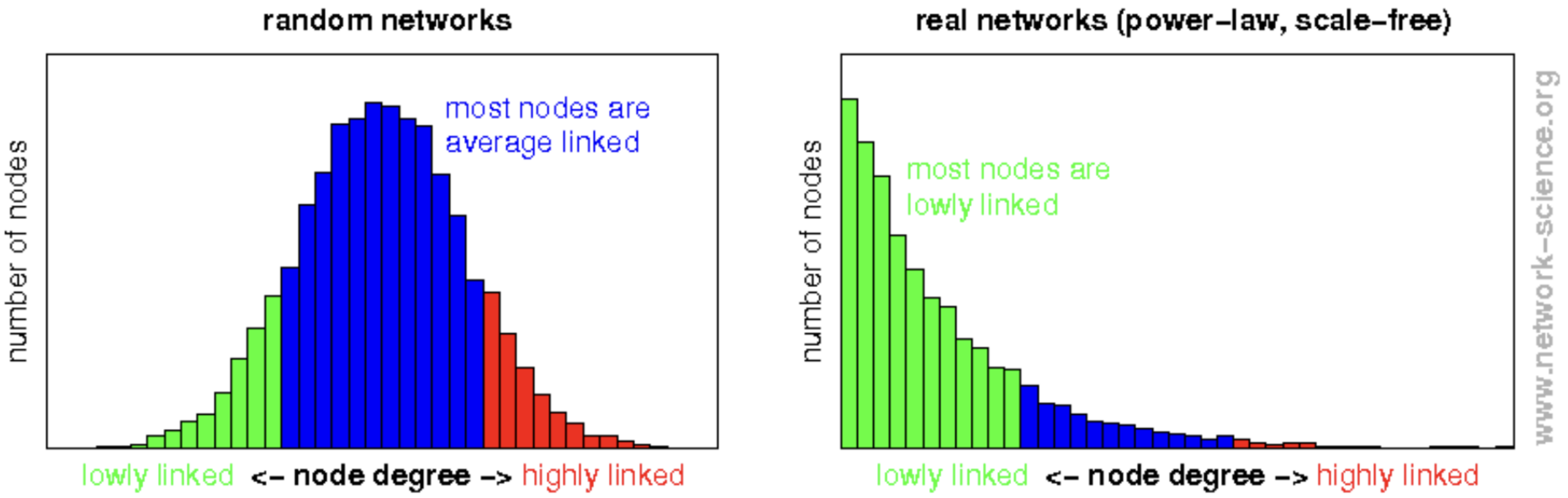
- Captures realistic feature of online networks

- Many nodes with few connections
- Few nodes with many connections

- This can create a power-law distribution



Power Law Distributions



- The central feature of the internet and computational social science is that we no longer can expect normal distributions.
- We have power-law distributions! Few actors control nearly all resources and

Other important Graphs

- **Caveman Graph**: Several tightly connected communities with few interconnecting edges.
- **Geometric Graph:** Nodes are connected if they are within a certain physical distance in space.
- **Grid Graph:** Nodes are arranged in a grid, each connected to its direct neighbors.
- **Star Graph:** One central node connected to all other nodes, with no other connections between peripheral nodes.

Diffusion in Networks

- If e.g., disease or information spreads in networks, does the network structure affect the outcome?


Erdos Renyi – Step 0
Barabasi Albert – Step 0
Watts Strogatz – Step 0
Caveman – Step 0
Grid – Step 0
Tree – Step 0
Star – Step 0
Geometric – Step 0

RESEARCH ARTICLE

Echo chambers and viral misinformation: Modeling fake news as complex contagion

Petter Törnberg

Published: September 20, 2018 • https://doi.org/10.1371/journal.pone.0203958

- Misinformation spreads like a virus, but not just any virus—it behaves more like a complex contagion.
- People often need to see misinformation multiple times from trusted people in their own group before they believe and spread it.
- Method: Network simulation with diffusion.
- Finding: Echo chambers make fake news spread faster and farther.
- Separate clusters counterintuitively intensify spread!

Structure of this Lecture

1. Questions you can answer with SNA
2. A brief history of social networks
3. Is there a theory of social networks?
4. Network basics: terms, foundations and examples
5. Visual Network Analysis
6. Community detection
7. Case studies: (a) Economic mobility (b) Zwarte Piet Debate on Twitter

Visual Network Analysis

- Visual network analysis involves the use of graphical representations to understand, interpret, and analyze the structure and characteristics of networks.

- This approach is particularly useful in network science, as it provides a way to intuitively comprehend complex relationships and patterns that might be difficult to discern through numerical data alone.

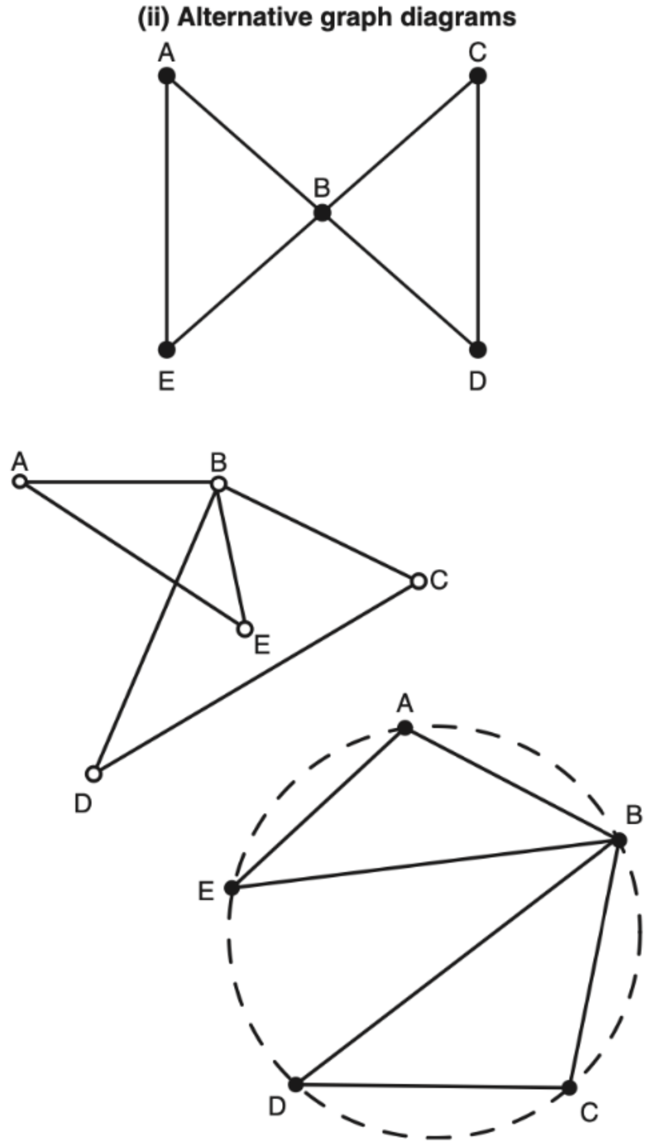


Figure 5.1 Alternative drawings of a graph

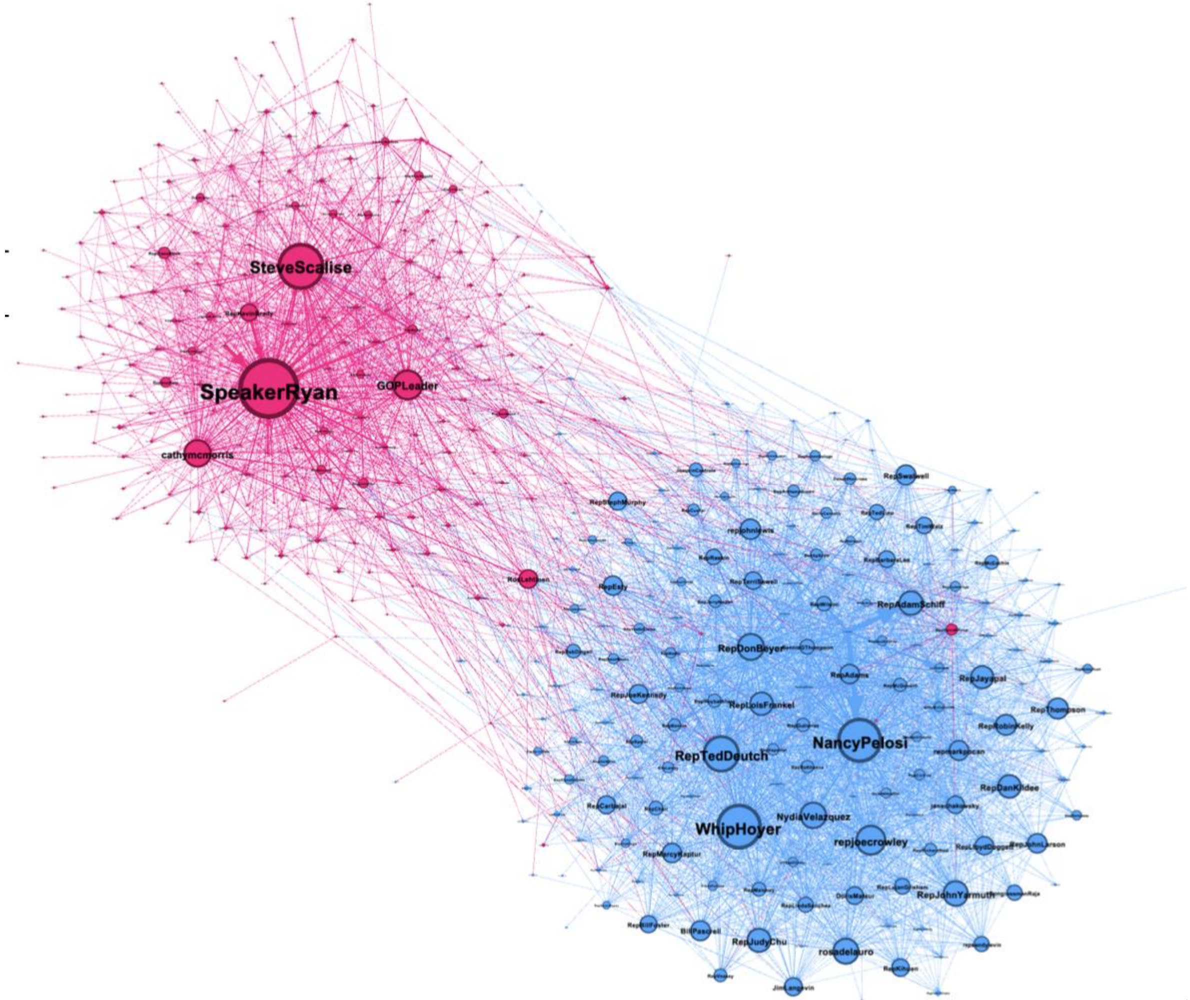
Layout Algorithms

- Layout algorithms in network analysis are used to determine the position of nodes and the arrangement of edges in a network visualization.

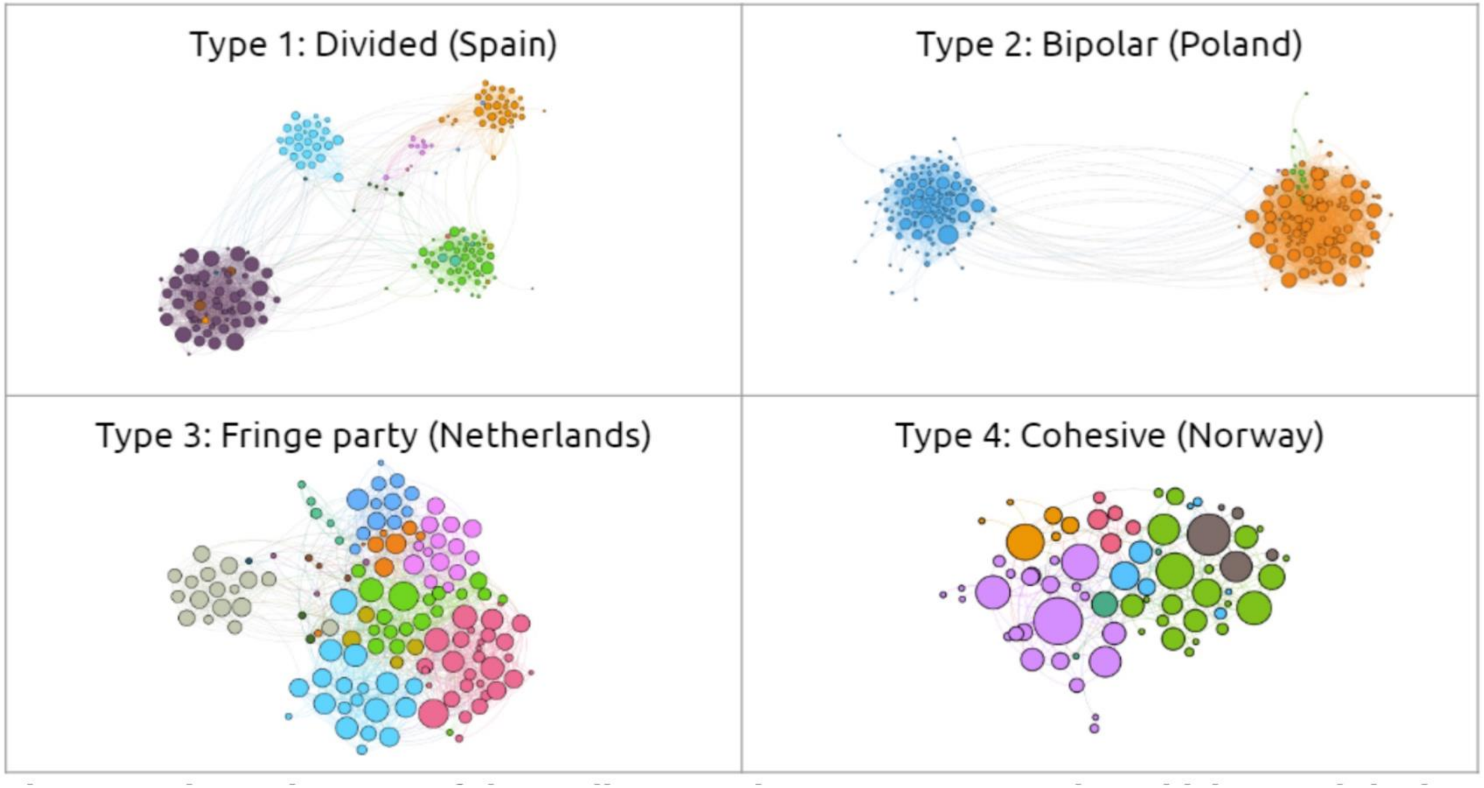
- **Force-Directed Layouts**: These simulate physical forces, treating edges as springs and nodes as objects repelling each other.
- **Circular Layouts**: These place nodes in a circle, which can be useful for highlighting cyclic structures and ensuring that all nodes are equally visible.
- **ForceAtlas 2** powerful and versatile force-directed layout.
- **Fruchterman-Reingold:** A type of force-directed algorithm that aims to produce aesthetically pleasing layouts by considering attractive forces (like springs) and repulsive forces (like charged particles).
- **Kamada-Kawai:** Another force-directed approach, focusing on accurately representing the distance between nodes in the layout, ideal for emphasizing the global structure of the network.
- **Hierarchical Layouts:** These are useful for directed networks (like organizational charts or decision trees), where nodes are arranged in layers, emphasizing the directionality of connections.
- **Grid Layouts**: These arrange nodes in a regular grid, which can be useful for networks with an inherent grid-like structure or when uniformity and alignment are important.

Example: Parliamentary Retweet Network

What can we say based on what we see here?



Example: How Can Polarization Look Like in Multiparty Systems?



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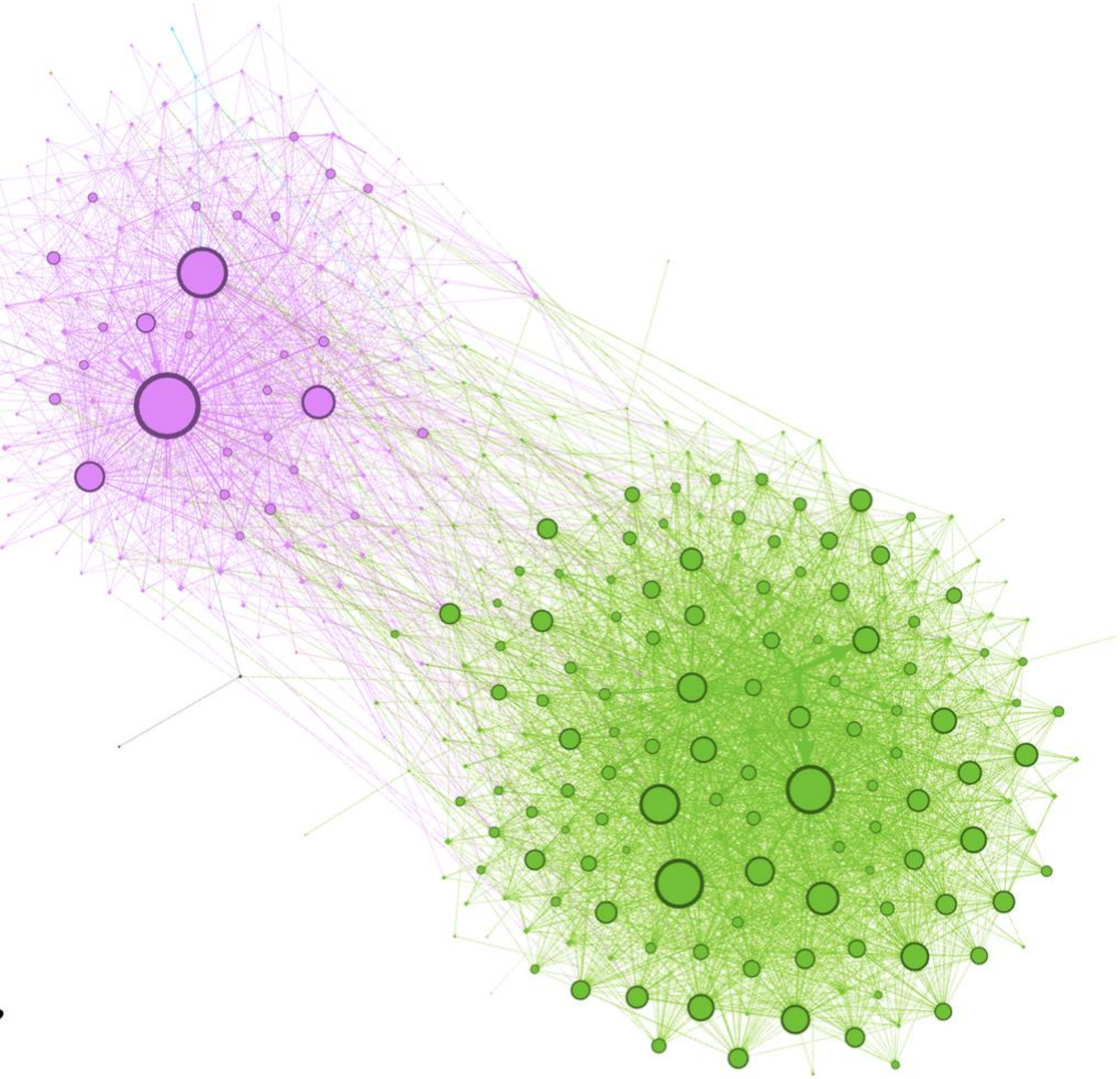
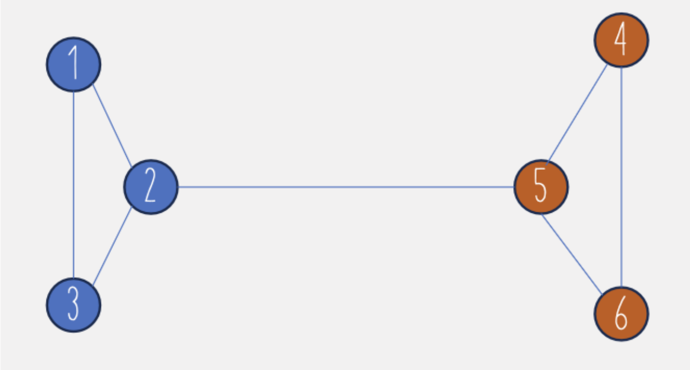
Community Detection

- Identify communities: groups of nodes that are more densely connected to each other than to the rest of the network.
- These communities often represent real-world structures like social circles, friend groups, or groups with common interests.
- Detecting these communities can be crucial for understanding the structure and function of social networks.



Running community detection on the U.S. House, it almost perfectly captures the two parties.

Calculating Modularity



$$Q = \frac{1}{2m} \sum_{ij} \left[A_{ij} - \frac{k_i k_j}{2m} \right] \delta(c_i, c_j)$$

- A_{ij} is the element of the adjacency matrix (1 if nodes i and j are connected, 0 otherwise).
- k_i and k_j are the degrees of nodes i and j.
- m is the total number of edges in the network.
- c_i and c_j are the communities of nodes i and j.
- $\delta(c_i, c_j)$ is the Kronecker delta function (1 if $c_i = c_j$, 0 otherwise).

- Basically, you calculate the number of edges between the communities and the edges within, and look at the fraction – while taking into consideration that some nodes are highly connected.

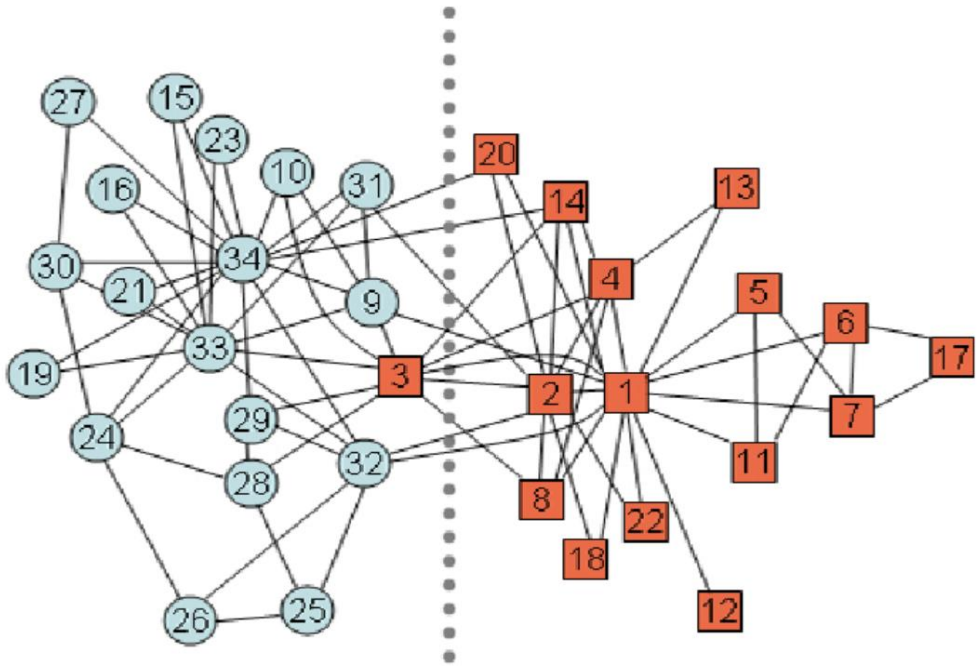
NOTE: Modularity unfortunately doesn't work well to compare large and small networks.

Community Detection Methods

- **Modularity-based** methods try to optimize modularity.
- **Clique Percolation Method (CPM):** find overlapping subgroups (cliques) within the network and then building communities from these cliques.
- **The Leiden Method (builds on Louvain)** is a popular approach: it's a hierarchical clustering algorithm that iteratively groups nodes to optimize the modularity score.

Example: Zachary’s Karate Club

- **Wayne W. Zachary** studied a katate club for a period of three years from 1970 to 1972
- Network: 34 members of the karate club, documenting links between pairs of members who interacted outside the club.
- During the study: a conflict developed bewteeent the club instructor and club administrator, becoming so severe that the club split up and join new clubs
- By analyzing the network Zachary managed to **predict almost exactly where the split would take place and which of the two clubs they would join.**
- "An Information Flow Model for Conflict and Fission in Small Groups" by Wayne W. Zachary



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Examples: Bringing the Concepts Together

Social Capital as a Network Question

nature

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nature > articles > article

Article | Open access | Published: 01 August 2022

Social capital I: measurement and associations with economic mobility

Raj Chetty, Matthew O. Jackson, Theresa Kuchler, Johannes Stroebel, Nathaniel Hendren, Robert B. Fluegge, Sara Gong, Federico Gonzalez, Armelle Grondin, Matthew Jacob, Drew Johnston, Martin Koenen, Eduardo Laguna-Muggenburg, Florian Mudekereza, Tom Rutter, Nicolaj Thor, Wilbur Townsend, Ruby Zhang, Mike Bailey, Pablo Barberá, Monica Bhole & Nils Wernerfelt

Nature **608**, 108–121 (2022) | Cite this article

- **Data:** Social networks of 72.2 million users of Facebook between 25 and 44 years;
  21 billion friendship ties in the United States
- Measure social capital at ZIP-code and country-level
- **Bonding capital →** strong within-group ties, often high clustering

  Bridging capital → cross-group ties, access to new information

Social Capital as a Network Question

- **Data:** Social networks of 72.2 million users of Facebook between 25 and 44 years; 21 billion friendship ties in the United States
- Measure social capital at ZIP-code and country-level

Social capital measured as three distinct network-related ideas:

- **Cross-type connectedness:** ties (friendships) across social groups, especially low-SES to high-SES ties **(bridging capital)**
  - Which concept from this lecture does this refer to?
  - What data do you need for this?

Social Capital as a Network Question

- **Data:** Social networks of 72.2 million users of Facebook between 25 and 44 years; 21 billion friendship ties in the United States
- Measure social capital at ZIP-code and country-level
- Social capital measured as three distinct network-related ideas:
- **Cross-type connectedness:** ties across social groups, especially low-SES to high-SES ties
- **Network cohesiveness**: degree to which friendship networks are clustered into cliques and whether friendships are supported by mutual friends (**bonding capital)**
  - What data do you need for this?

Social Capital as a Network Question

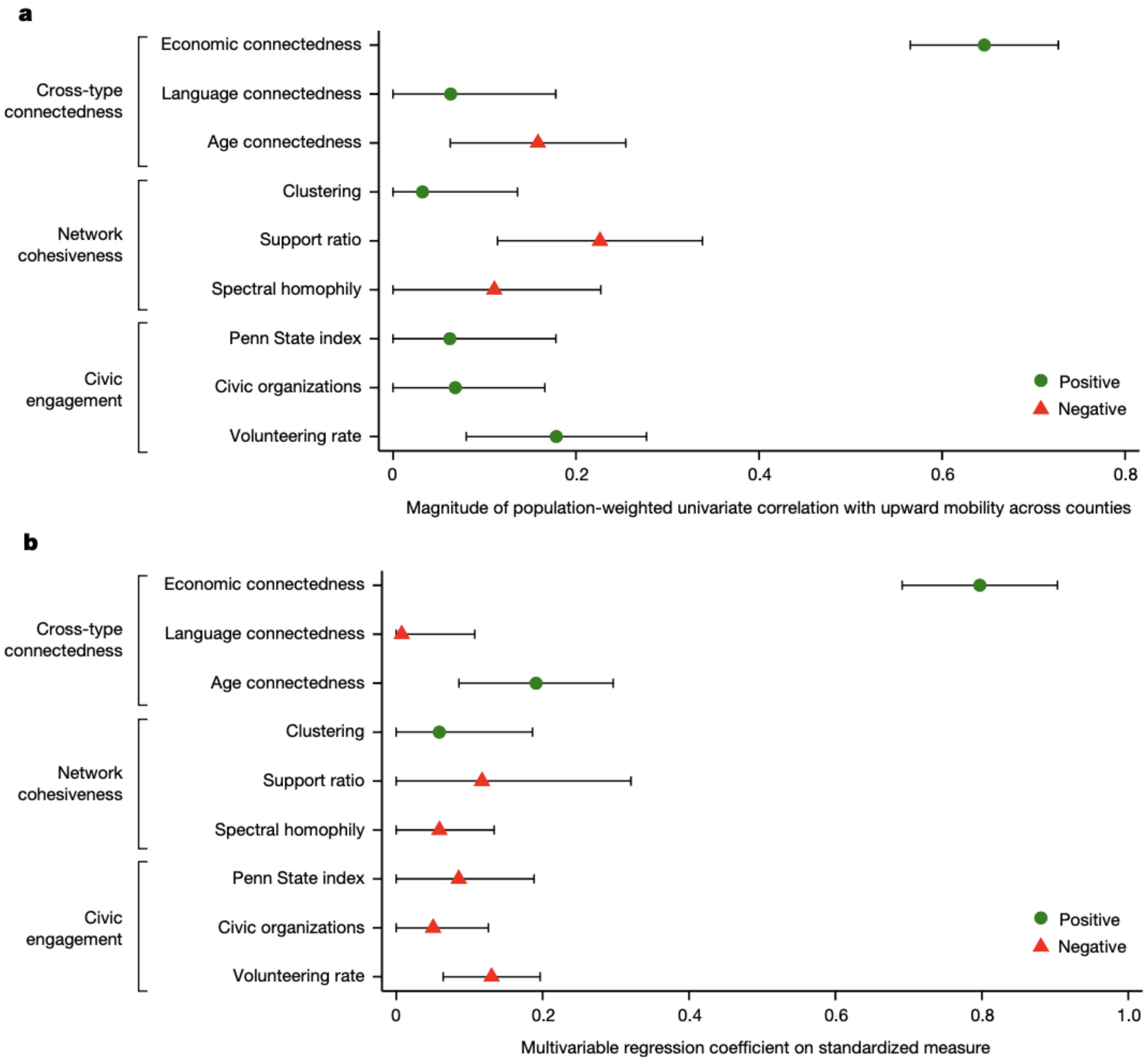
- **Data:** Social networks of 72.2 million users of Facebook between 25 and 44 years; 21 billion friendship ties in the United States
- Measure social capital at ZIP-code and country-level
- Social capital measured as three distinct network-related ideas:
- **Cross-type connectedness:** ties across social groups, especially low-SES to high-SES ties
- **Network cohesiveness**: clustering, support ratio, fragmentation
- **Civic engagement:** volunteering and civic organization membership (non-network)
- The authors focus on: **Economic connectedness (EC)**
  the extent to which people with low and high SES are friends with each other.

Findings

- **Friendship is strongly homophilous by socioeconomic status:**
  As a person's SES rises, the average SES of their friends also rises.
  Among below-median-SES individuals, only **38.8%** of friends are above-median SES; among above-median-SES individuals, the number is **70.6%.**

- **Economic connectedness** is the standout predictor of upward mobility

- **By contrast, clustering, support ratio, and civic engagement are much weaker and less stable predictors.**



Implications:

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- Bridging ties across class lines might matter more for economic mobility than purely local cohesion.

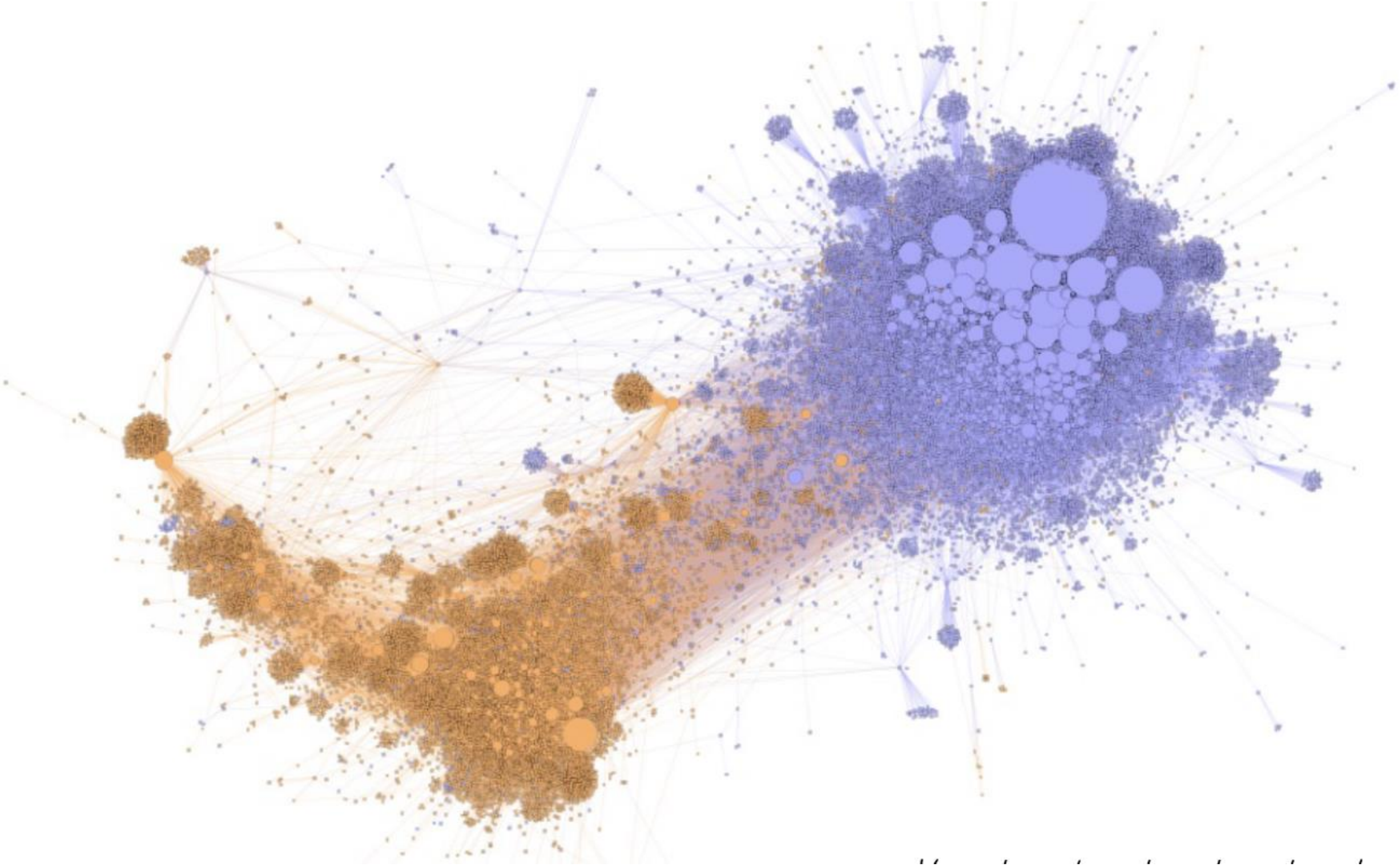
- For “getting ahead,” bridging capital (brokerage) might matter more than bonding alone (community)

→ Implications of this study: If low-SES children grew up in counties with EC like that of the average high-SES child, their adult incomes would rise by about 20%

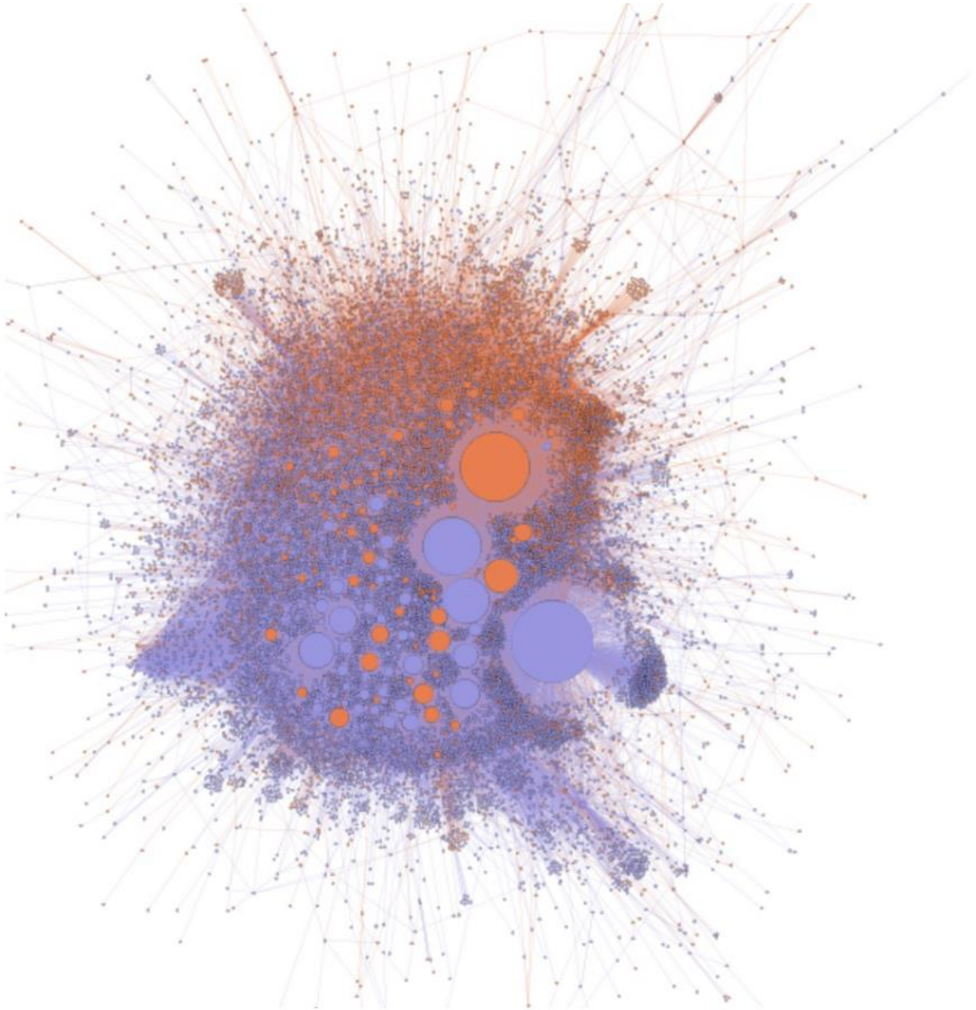
Example 2: The zwarte piet debate

Case study:
- Lightning-rod for Dutch culture wars
- Tweets from December 2017 to May 2019.
- 430,000 tweets from 81,700 users, with 296,881 mentions between users.

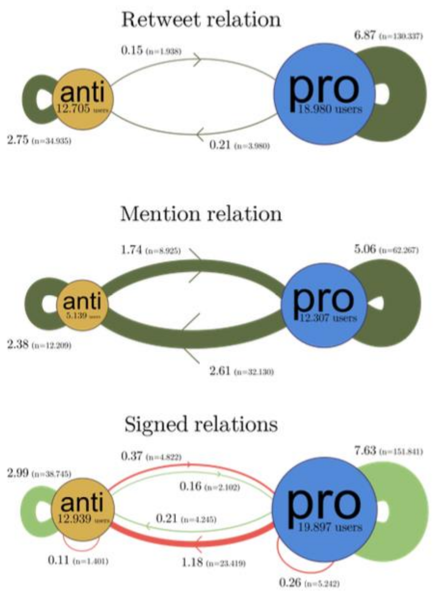
Retweets

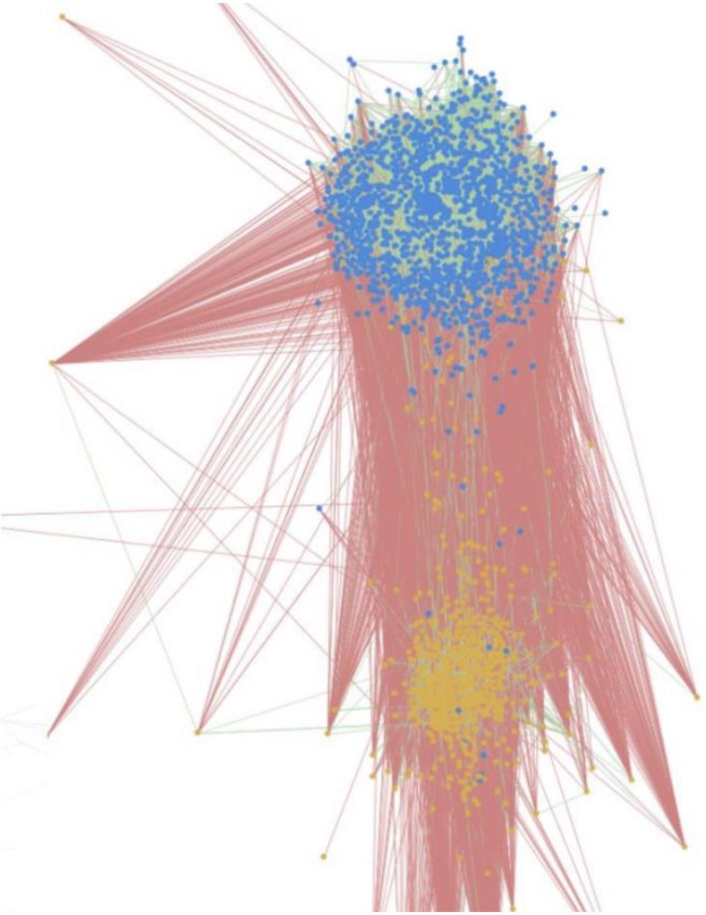


Mentions



- Trained a machine learning algorithm to identify whether links were positive or negative.
- The groups are defined as much by external attacks as by internal support
- Conversation is not rational exchange, but predominantly conflictual and often uncivil – ‘outgroup derogation’





Keuchenius, A., Törnberg, P., & Uitermark, J. (2021). Why it is important to consider negative ties when studying polarized debates: A signed network analysis of a Dutch cultural controversy on Twitter. PloS one, 16(8).

Remaining Questions?